

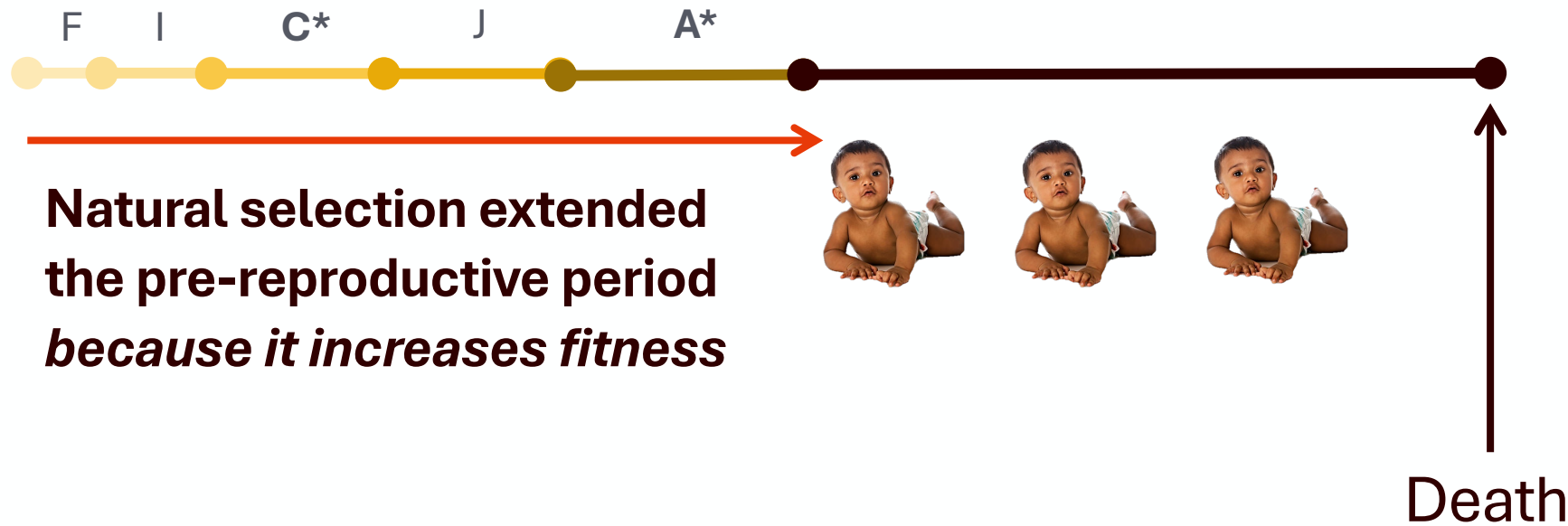


Ageing, senescence, and mortality

Dr. Laura van Holstein

lavanho@emory.edu

ANT215



The point of life is to convert energy to offspring

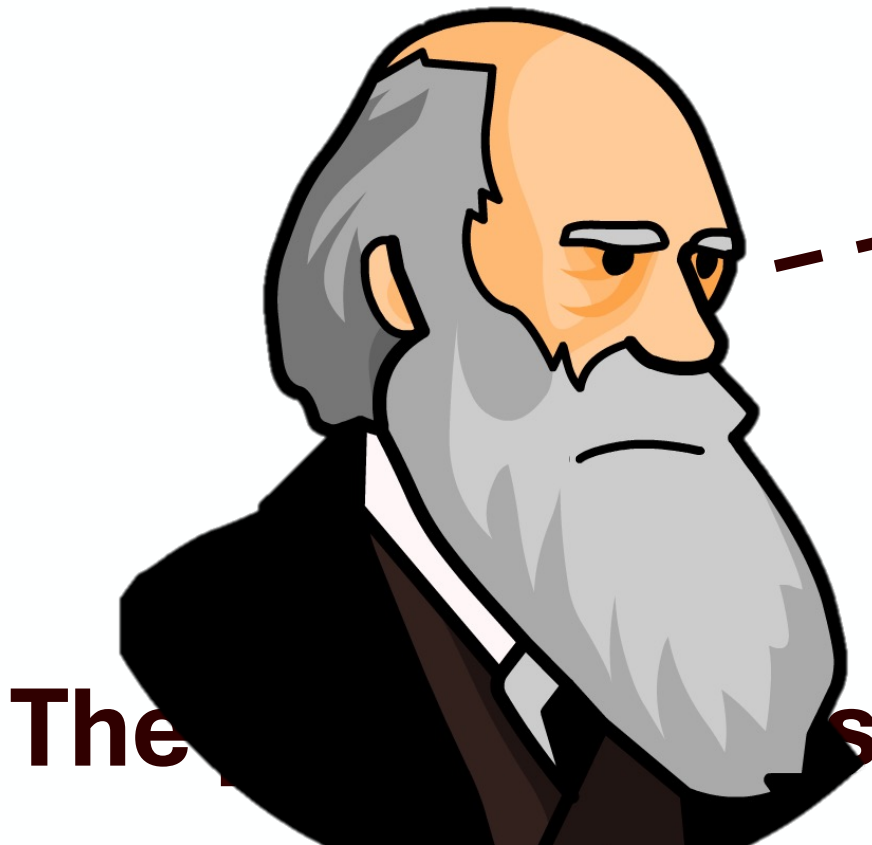


Natural selection extended
the pre-reproductive period
because it increases fitness



Death

Why do organisms die?
Is death adaptive?



The **cost** of converting energy to offspring



Species

Populations

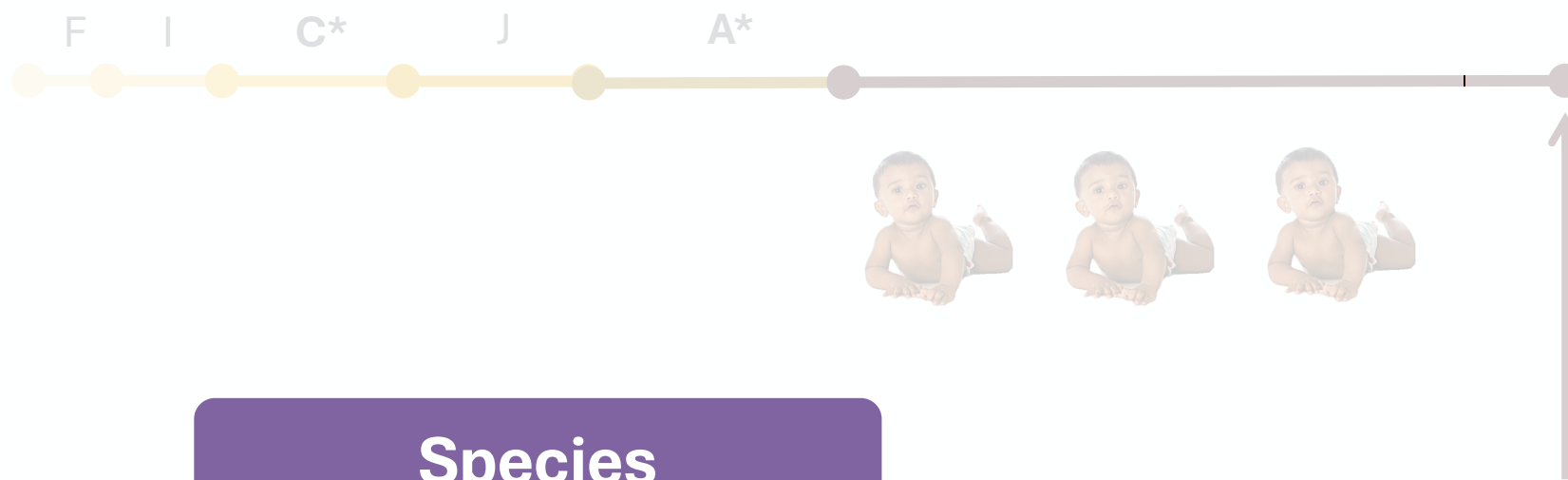
Individuals

Genes

Death

Why do organisms die?
Is death adaptive?

The **goal** is to convert energy to offspring



Death

Species

Populations

Individuals

Genes

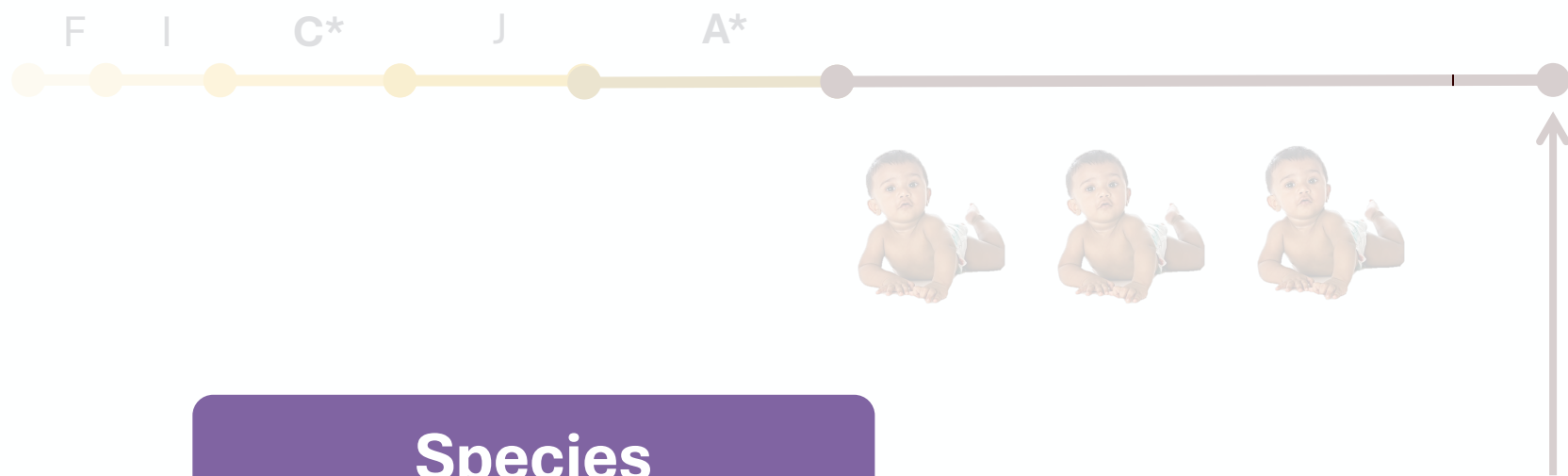
Why do organisms die?

Why do organisms live?



But... continuing to have babies is better than not doing so!

The costs of living



Death

Why do organisms die?

Why do organisms live?

- Species
- Populations
- Individuals
- Genes



But... continuing to have babies is better than not doing so!

The costs of living

**Weisman (1899):**

Although ageing is bad for individuals, it **benefits groups** by limiting population density and therefore allowing more space and resources for **younger, fitter, more energetic individuals**

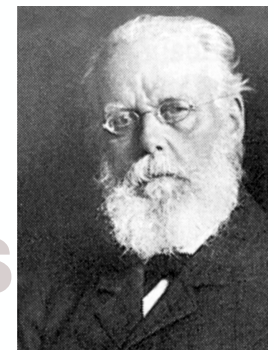
Species**Populations**

Individuals

Genes

But...

- Individual-level selection usually supersedes group selection!
- In the wild, a “death programme” would only concern a small number of individuals who would have survived
- In most species, age-specific mortality rates increase steadily while in the programmed death scenario a ‘wall of death’ would be expected at a particular age



The **task** is to convert energy to offs



Species

Populations

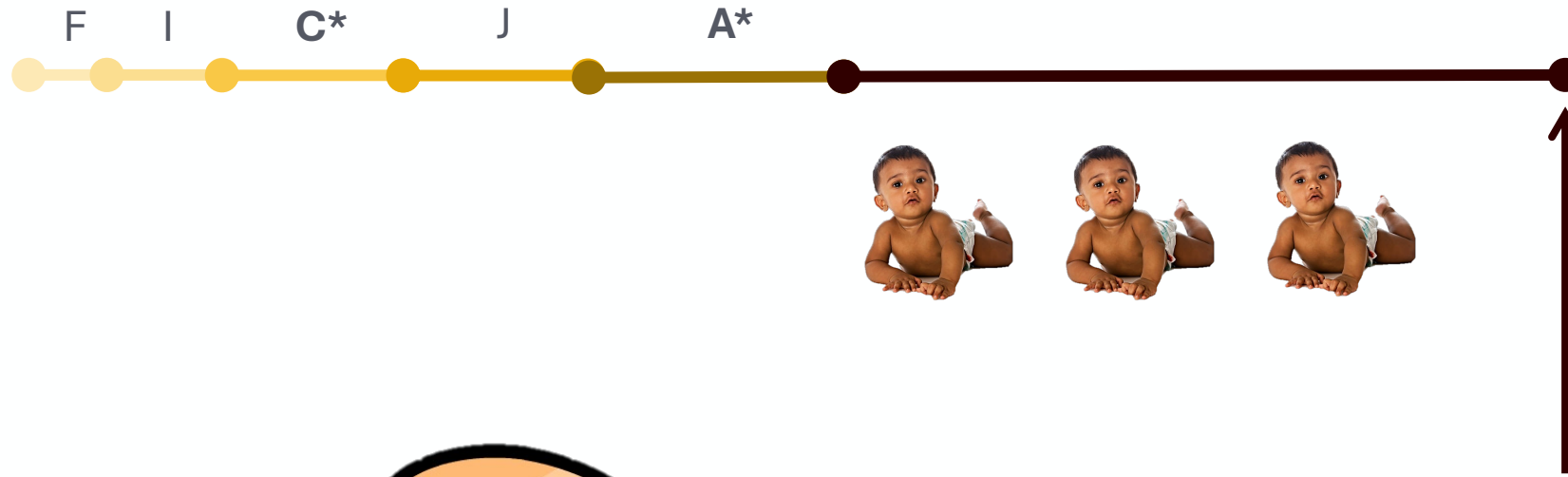
Individuals

Genes

Death

Why do organisms die?
Is death adaptive?

The **task** is to convert energy to offspring



Death

Why do organisms die?
Death is not adaptive.

The  needs to convert energy to offspring



Death

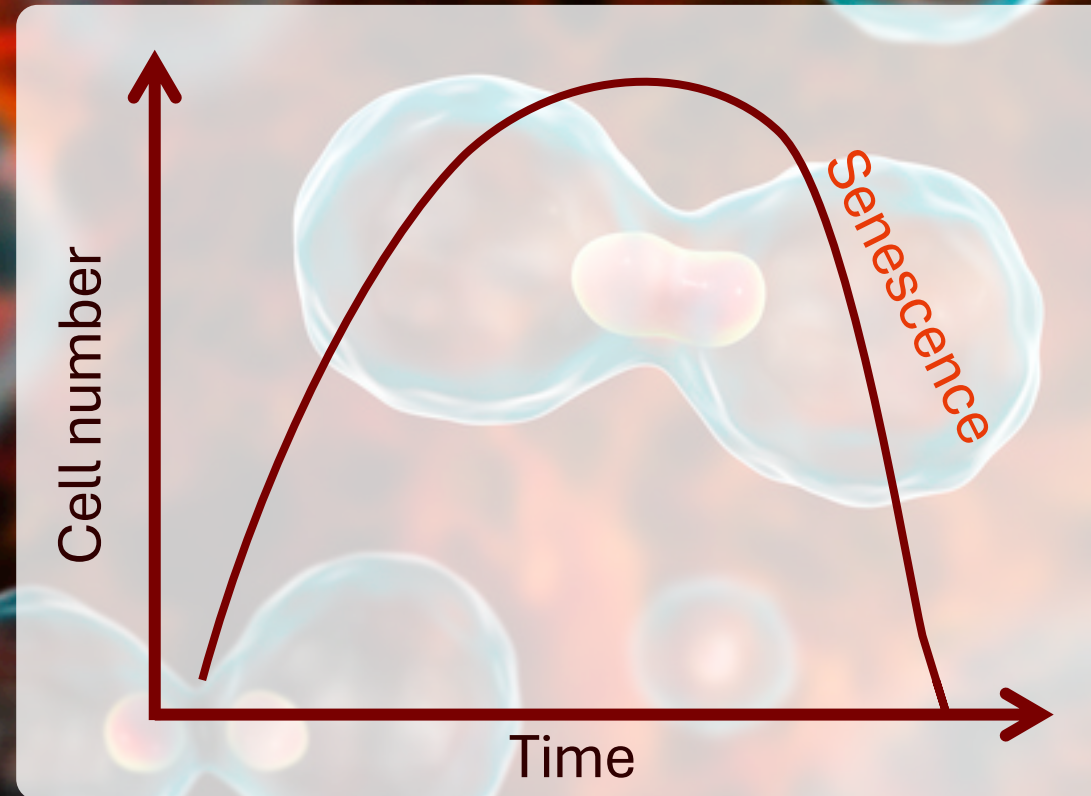
How do organisms die?

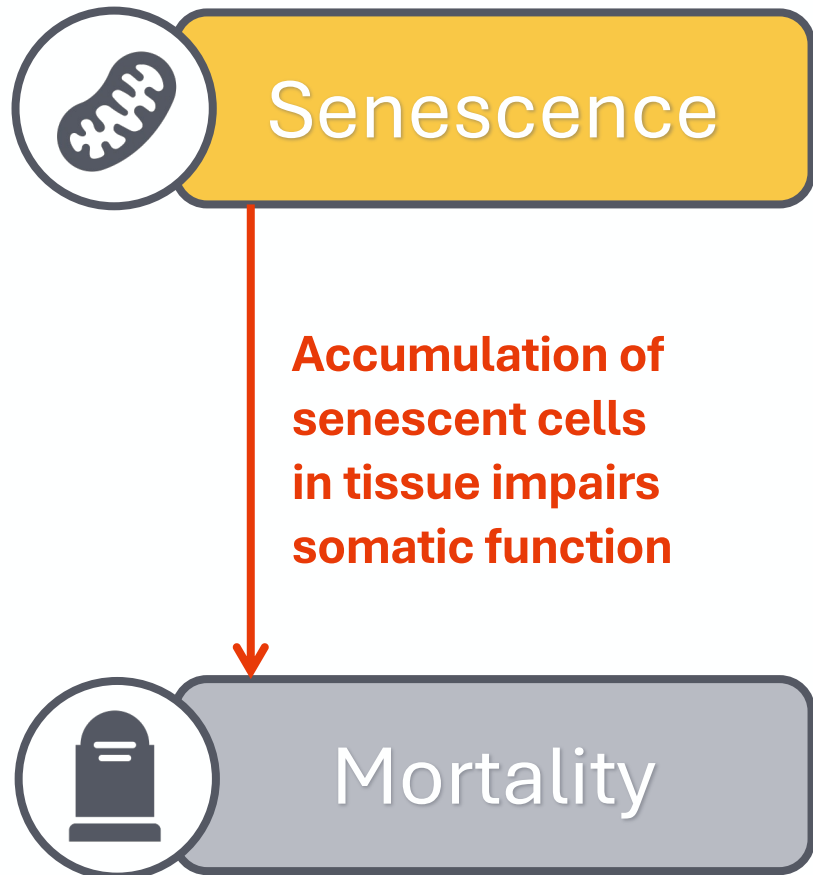


The goal is to convert energy to offspring

Hayflick limit

Normal human cells can divide 40-60 times





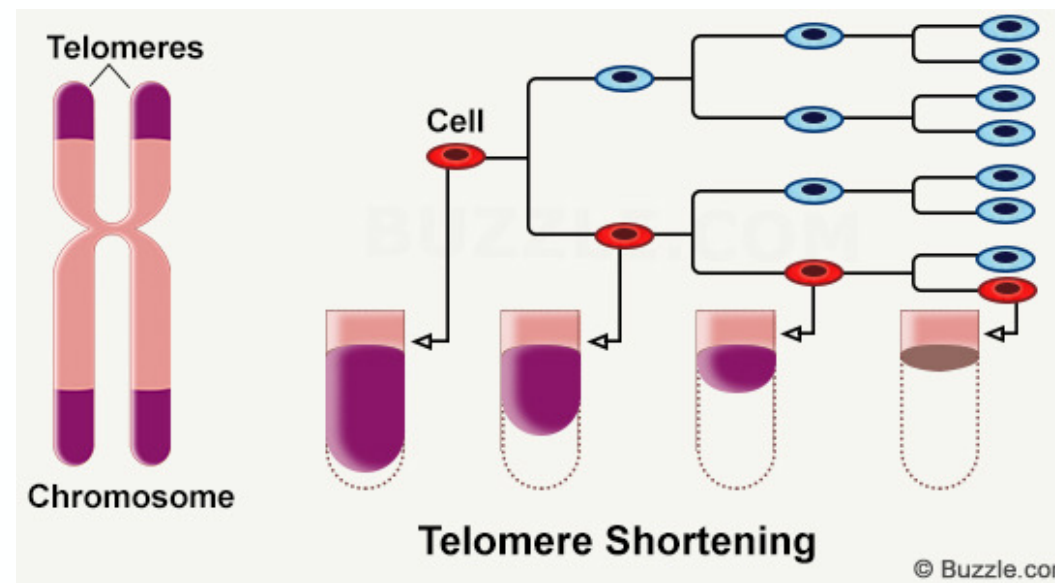
Hayflick limit:

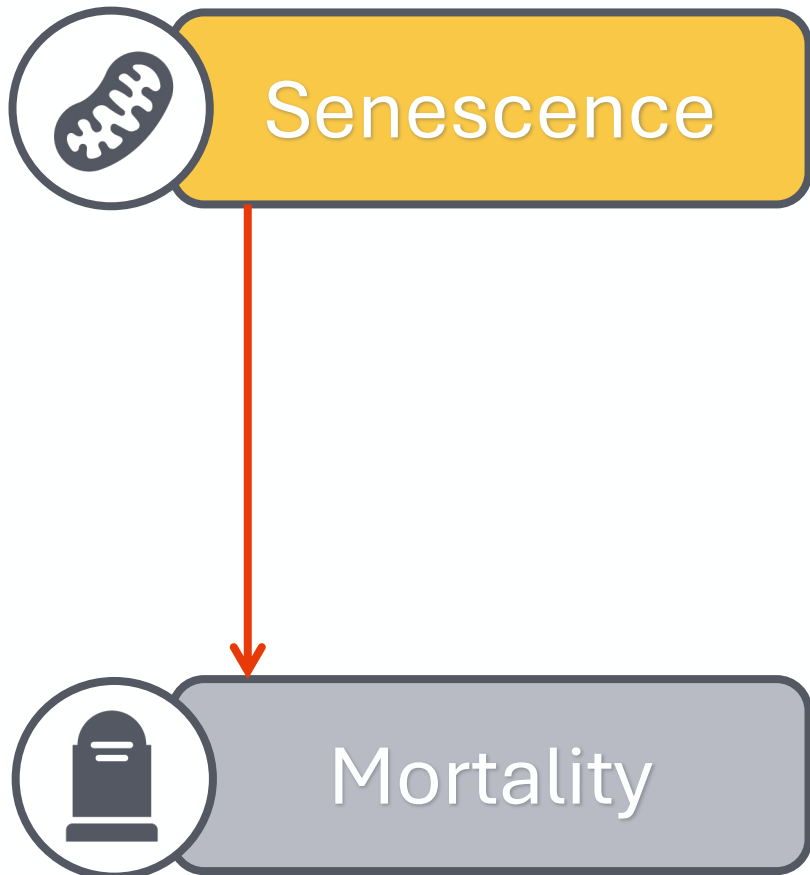
the number of times a normal cell population can divide before entering the replicative senescence phase

Replicative senescence:

the inability of cells to divide. Comes with

- expressing certain proteins
- releasing inflammatory molecules
- loosening tightly packed regions of DNA known as heterochromatin



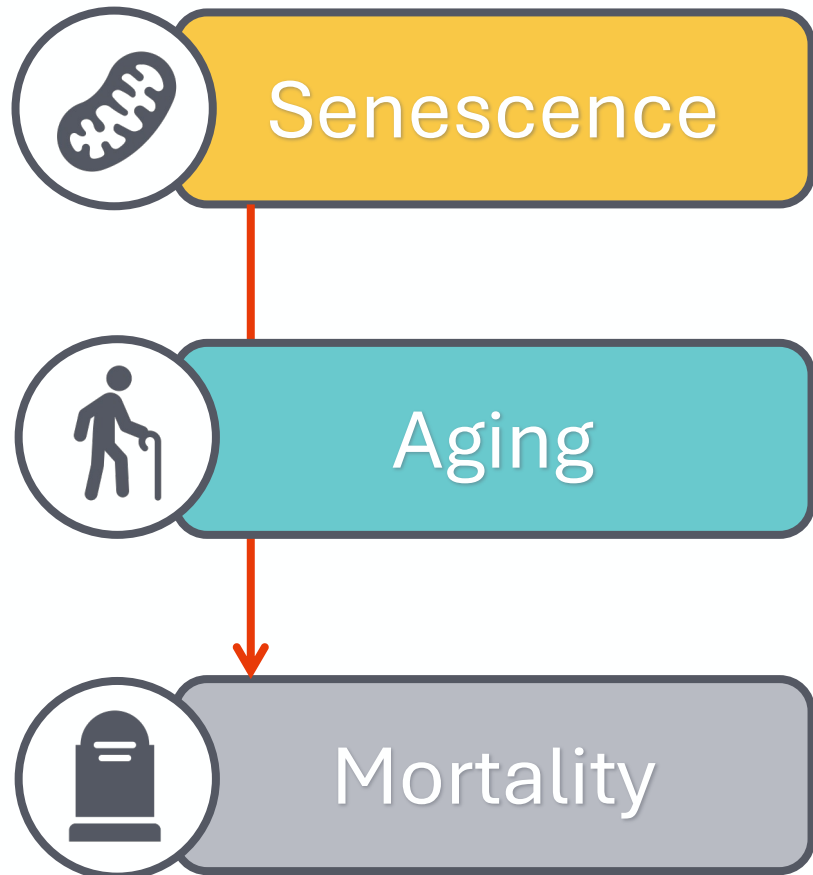


Crews and Ice (2012):

“Senescence is **progressive degeneration** following a period of development and attainment of maximum reproductive potential that **leads to an increased probability of mortality**”

It includes:

- **Intrinsic biological processes that lead to declines in cellular function**
- An increased probability of death
- Reduction of reproductive potential over the lifespan

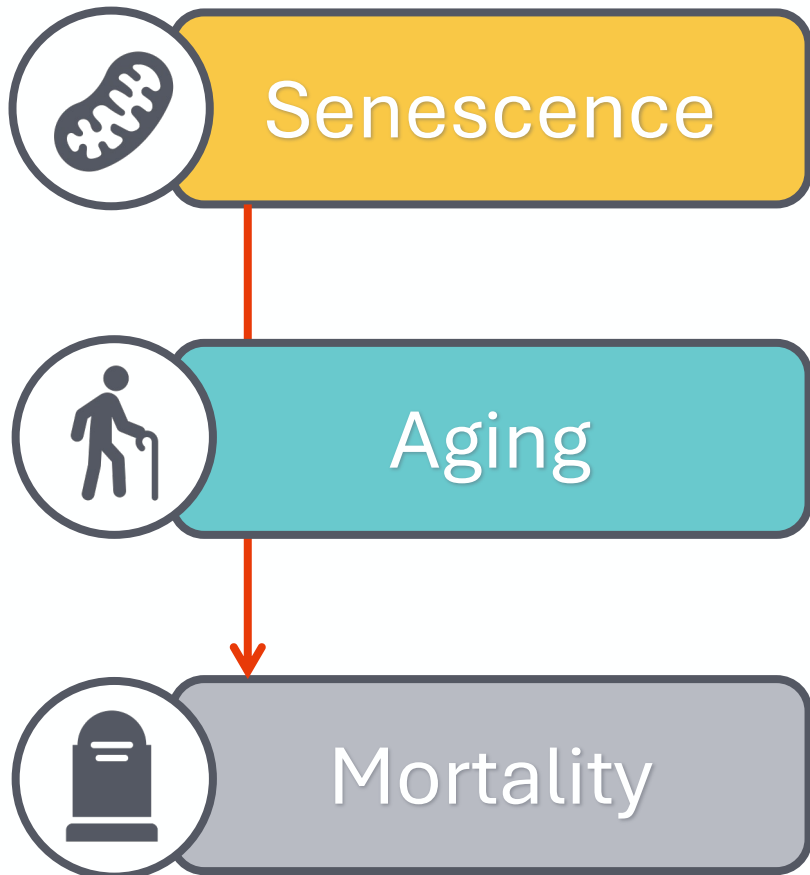


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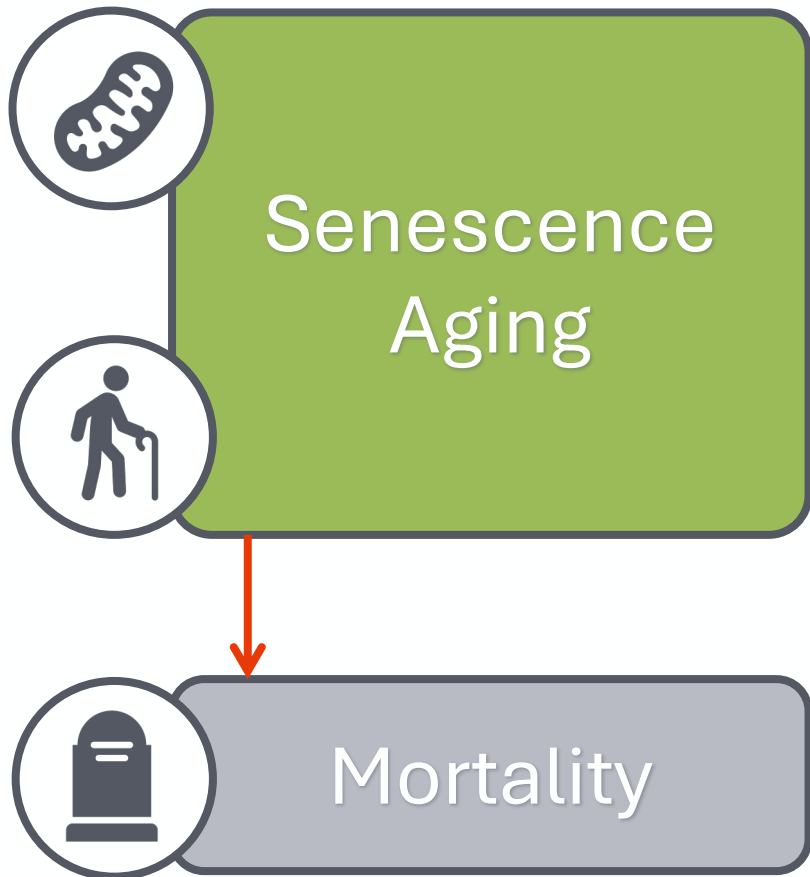
- **Intrinsic biological processes that lead to declines in cellular function**
- **An increased probability of death**
- **Reduction of reproductive potential over the lifespan**



Crews and Ice (2012):

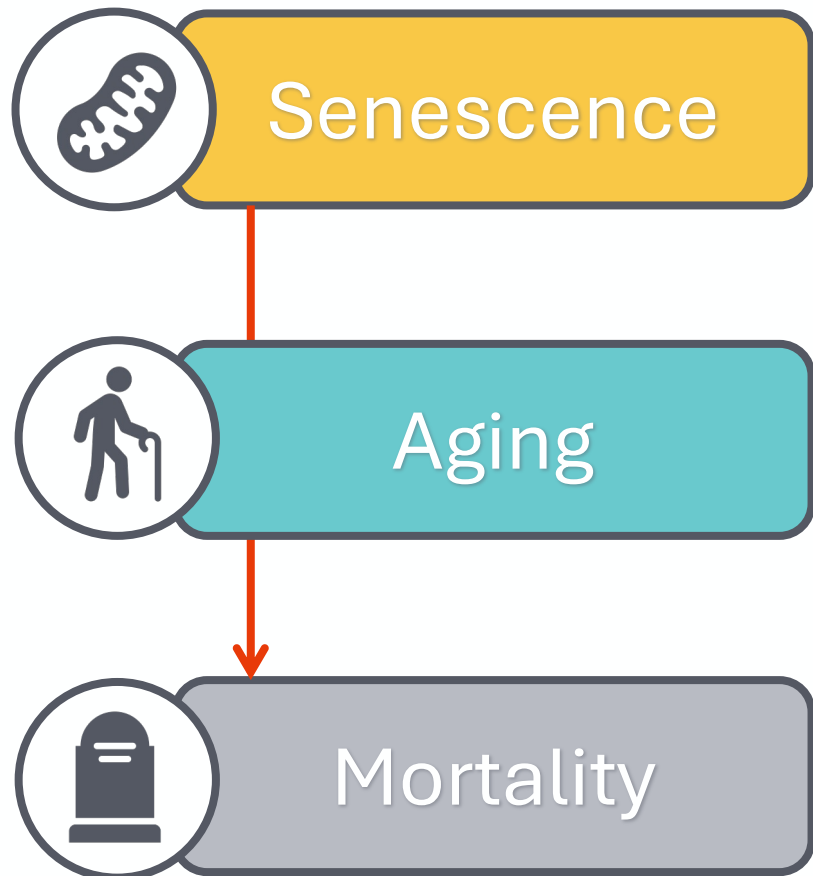
Ageing: the outward **phenotypic signs** of senescence:

- Increased susceptibility to infectious and chronic diseases
- Loss of resistance to external and internal stressors
- Inabilities to maintain and repair somatic systems
- Progressive increase in frailty



Maklakov and Chapman (2019):

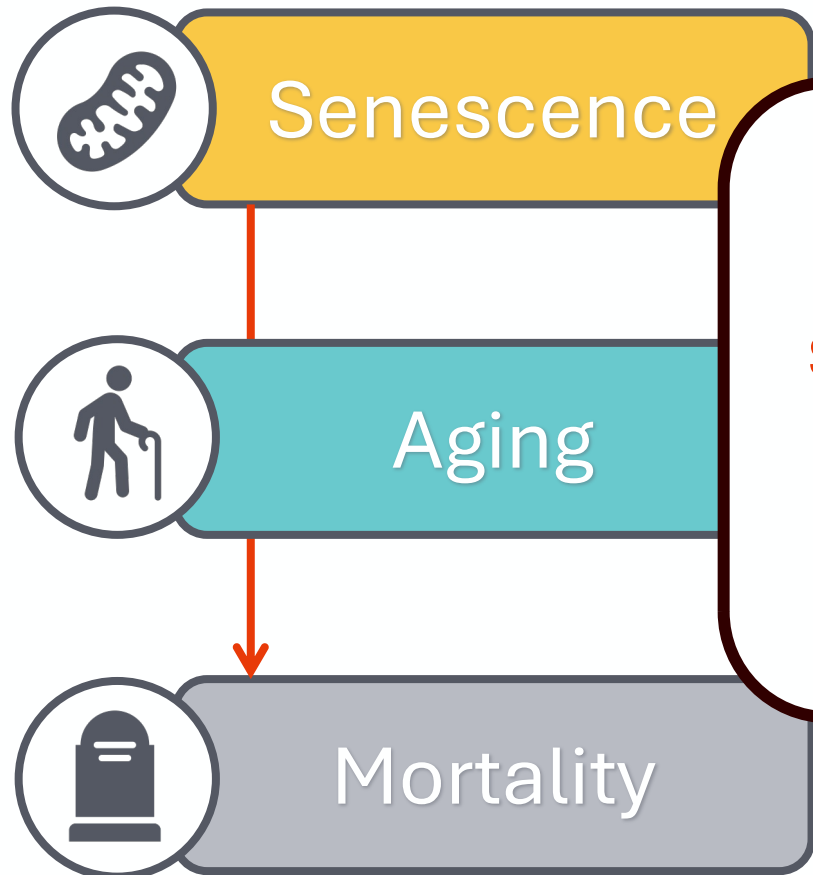
“**Aging, or senescence**, is a physiological deterioration of an organism with advancing age, which reduces reproductive performance and increases the probability of death...”



Some of the processes that underlie senescence and ageing:

1. Telomere attrition: telomeres shortening
2. Genomic instability: Excessive DNA damage or insufficient DNA repair
3. Epigenetic alterations: Alterations in the methylation of DNA or acetylation and methylation of histones, as well as of other chromatin-associated proteins
4. Loss of proteostasis: Failure to refold or degrade unfolded proteins
5. Deregulated nutrient sensing
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7. Stem cell exhaustion
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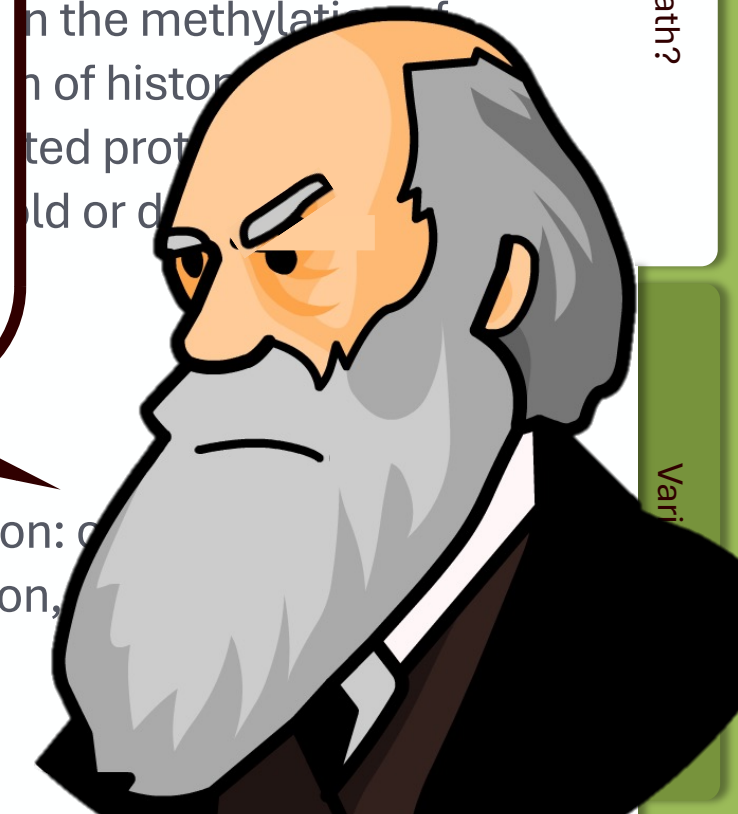


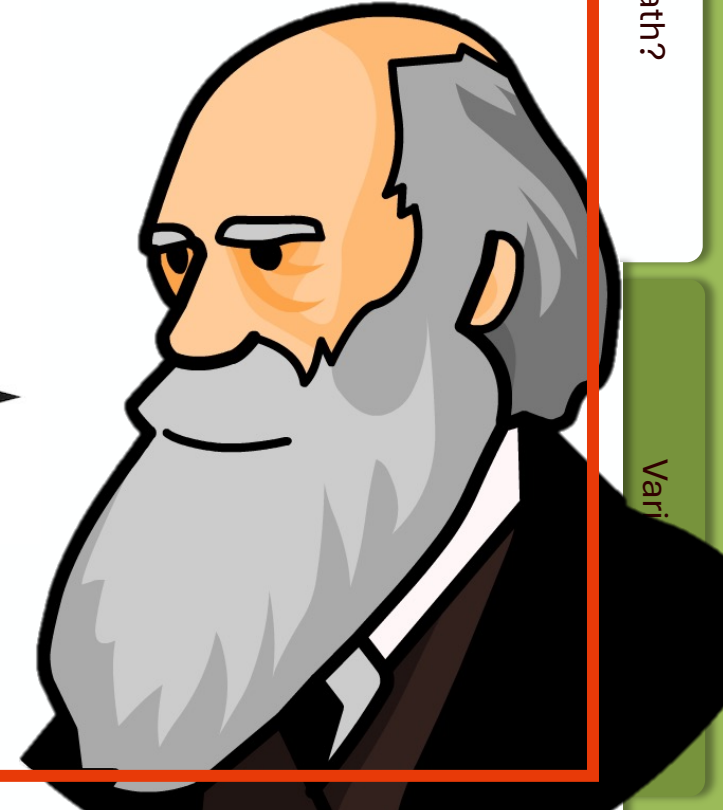
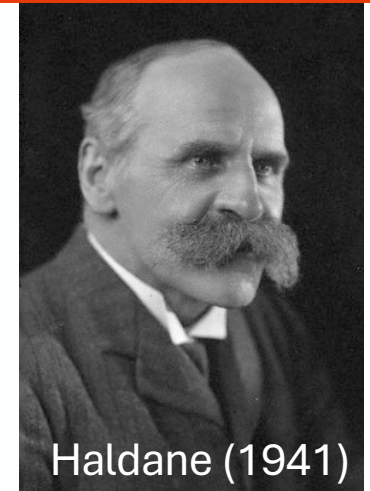
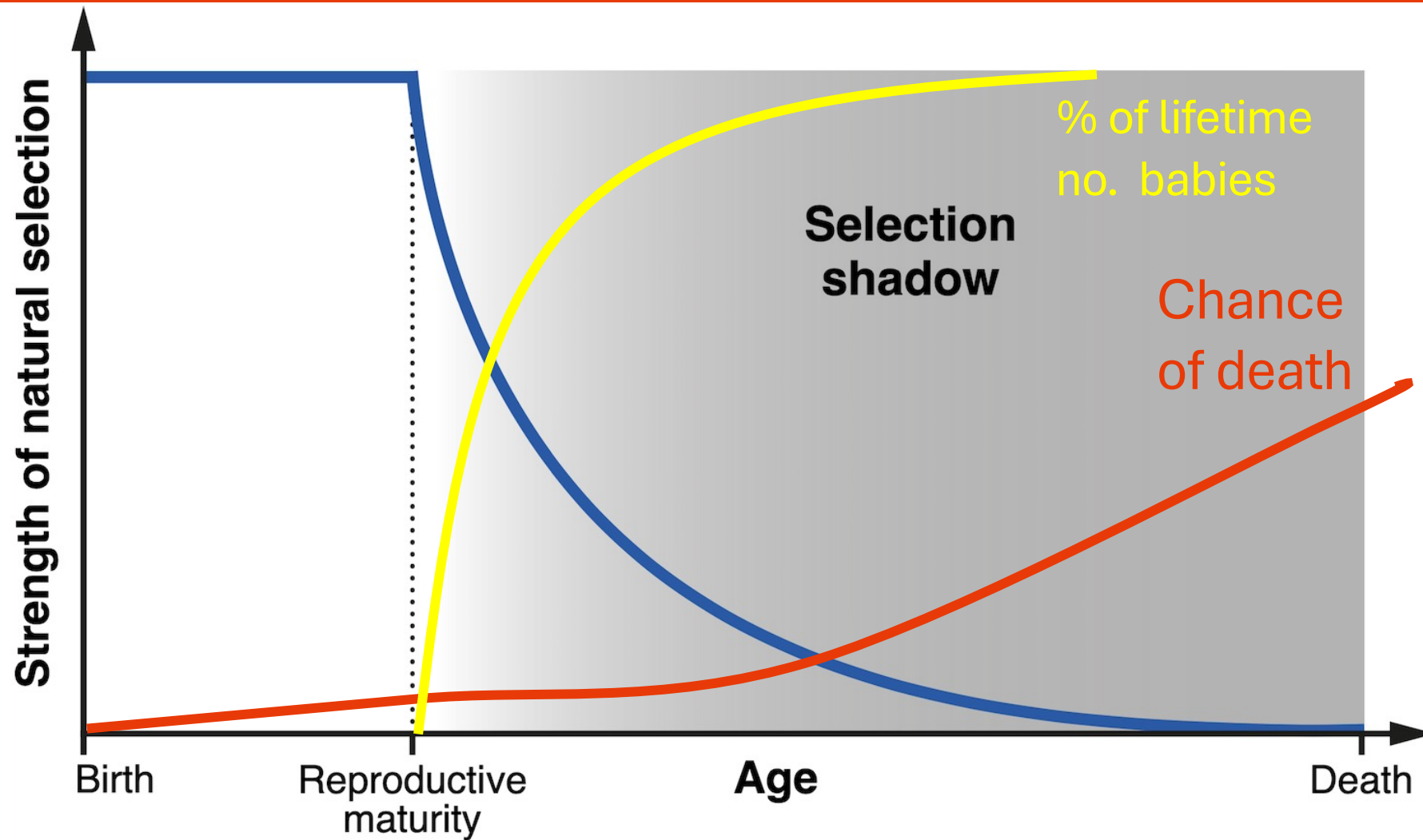
Why has natural selection not selected against these processes?!

- 6. Mitochondrial dysfunction
- 7. Stem cell exhaustion
- 8. Altered intercellular communication: decreased level of intercellular communication, neuroendocrine or neuronal

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...NA damage or

...n the methylation of
...n of histone
...ted proteins
...ld or d

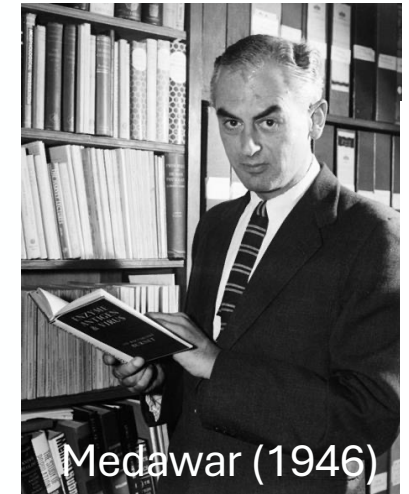
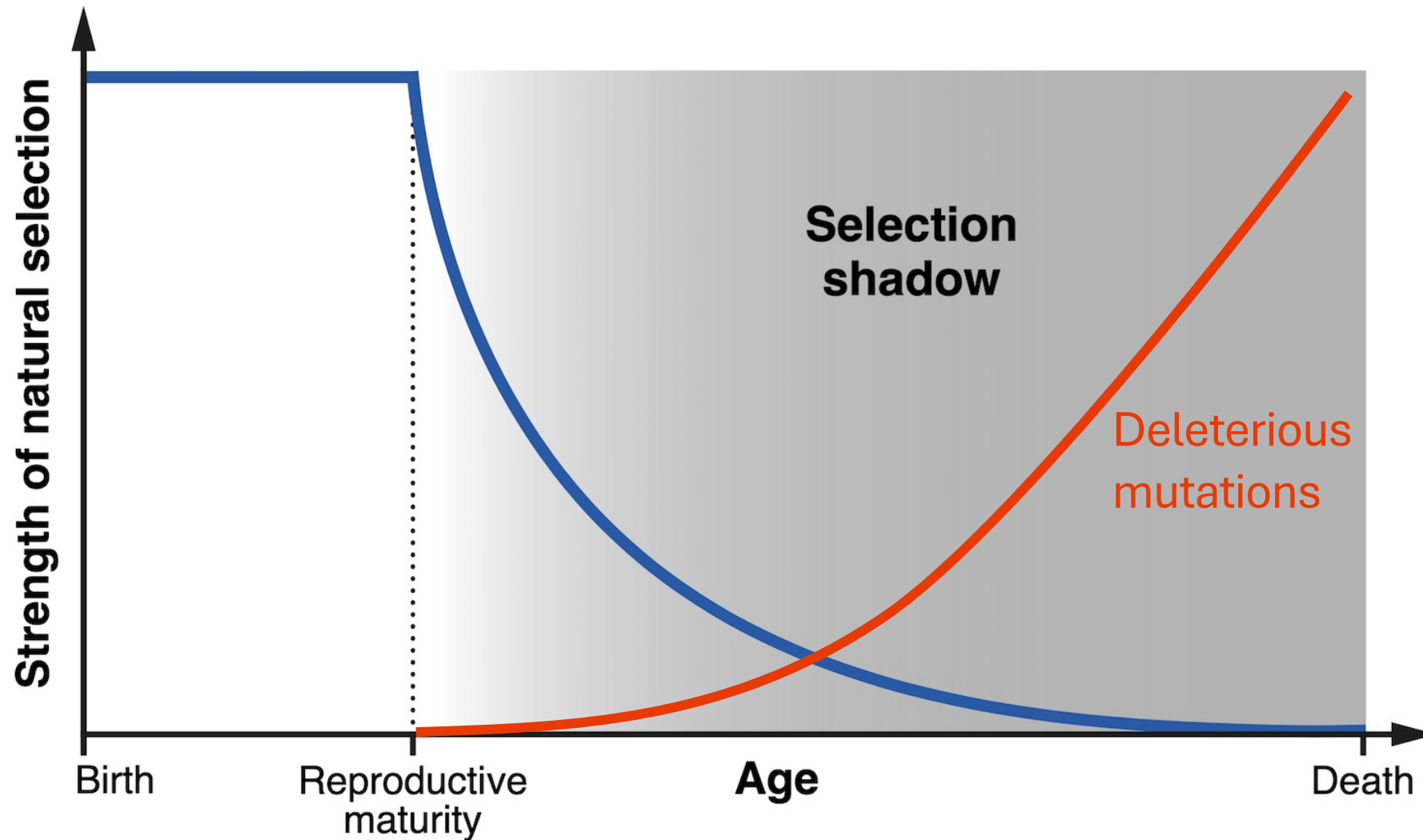




Intro

How death?

Vari



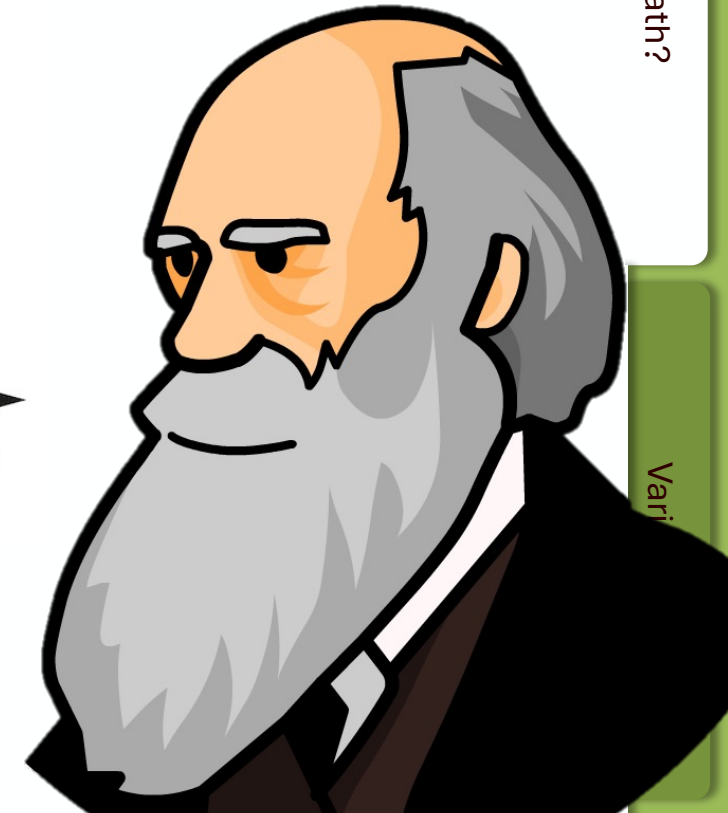
Intro

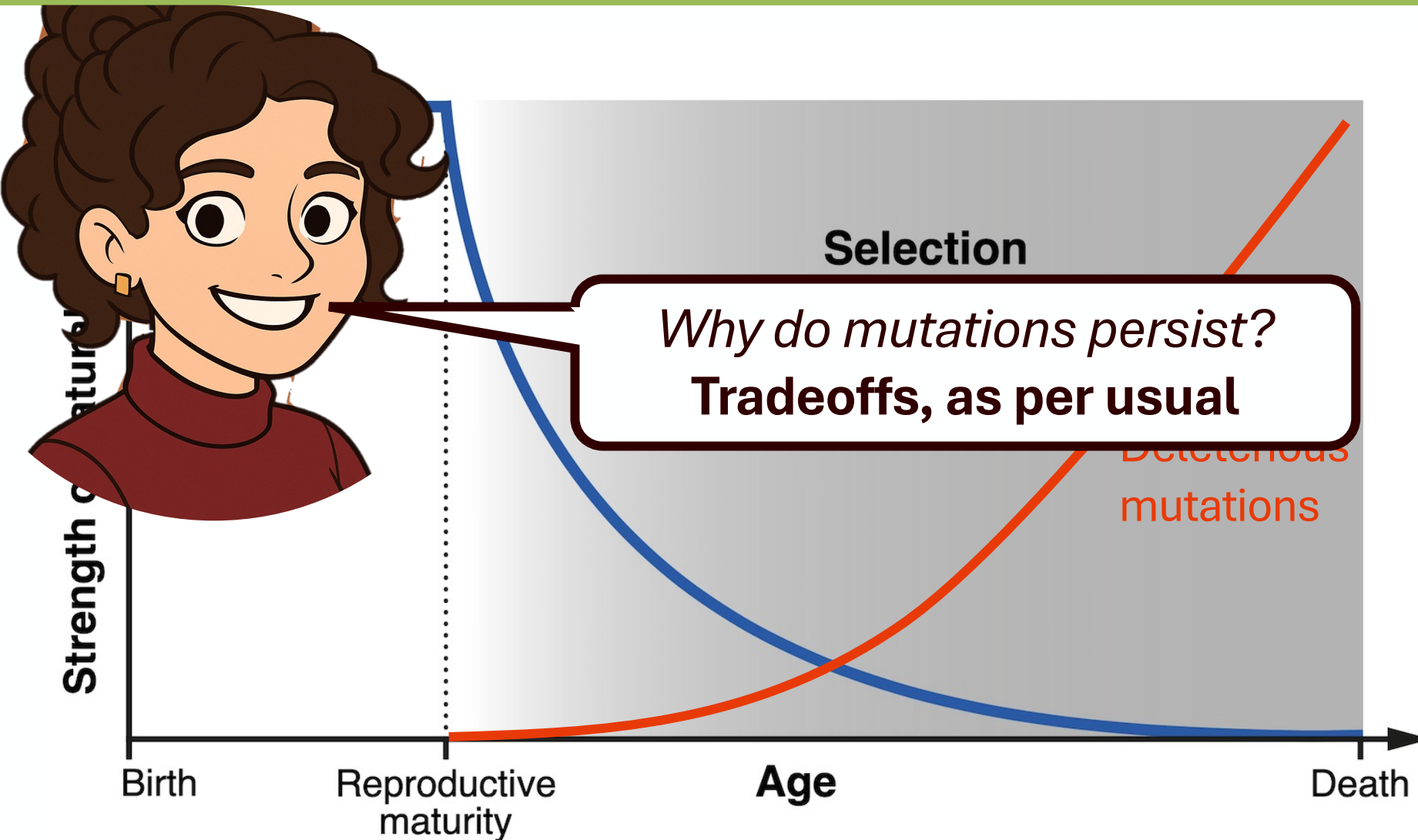
How death?

Vari

How does this lead to senescence?

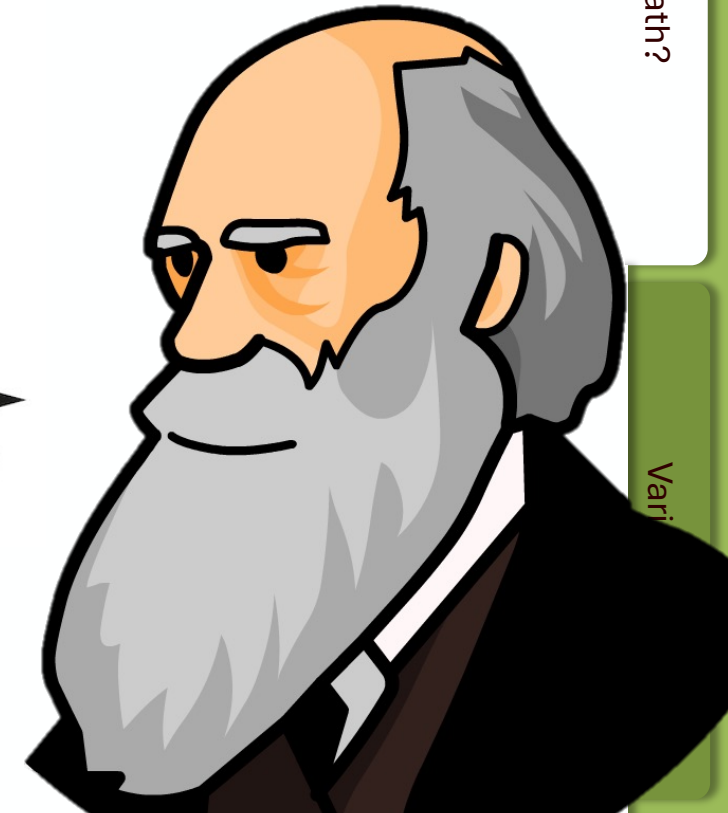
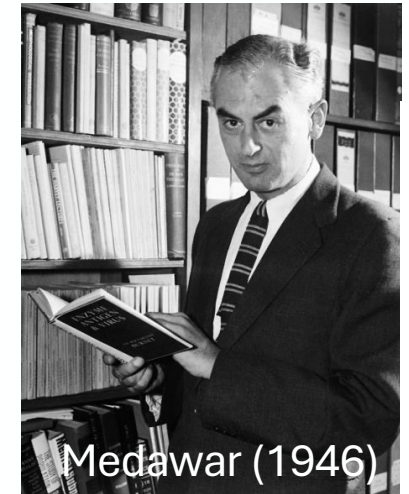
because strength of selection decreases, no mechanisms to counter mutations evolves





How does this lead to senescence?

because strength of selection decreases, no mechanisms to counter mutations evolves





Why do mutations persist?
Tradeoffs, as per usual

Maintenance

Reproduction



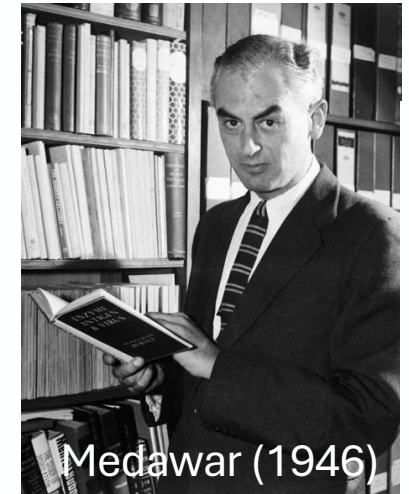
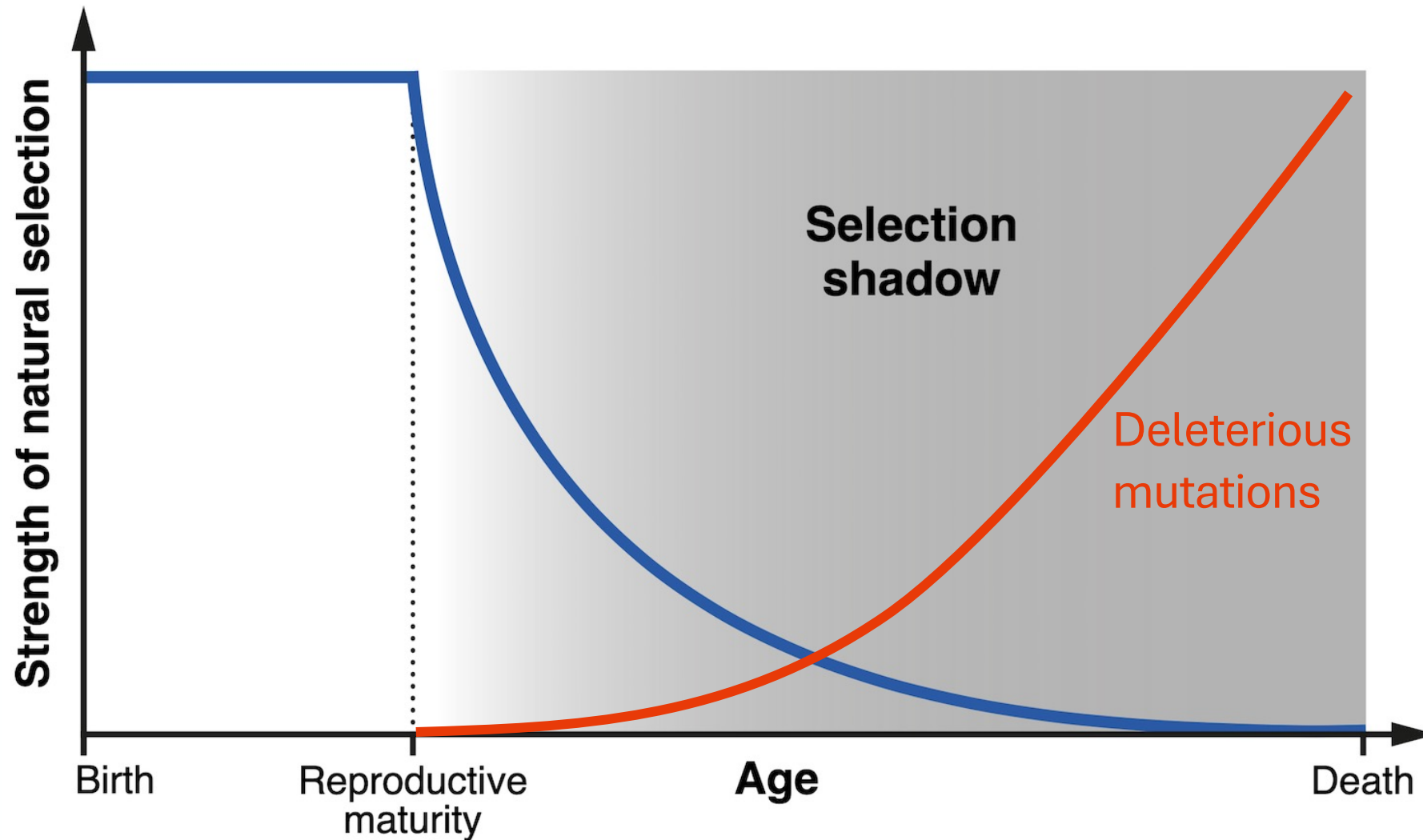


Why do mutations persist?
Tradeoffs, as per usual

Maintenance



*Energy not used
for cellular repair*



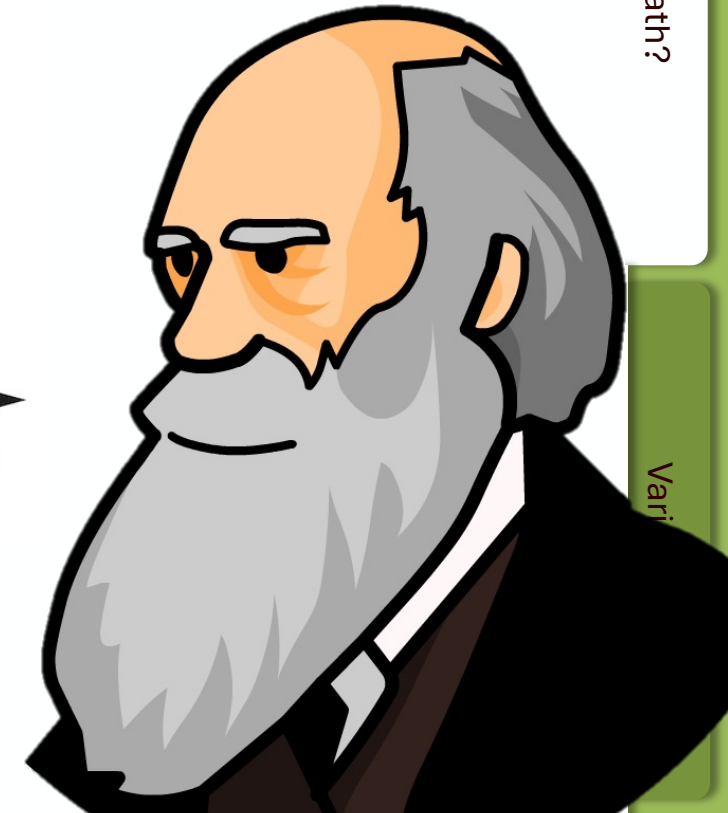
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How death?

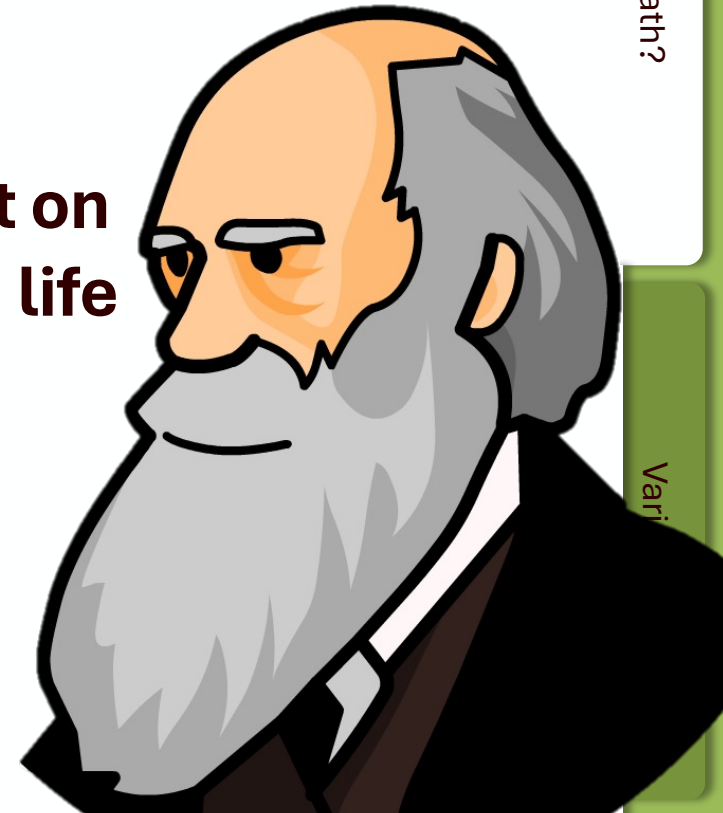
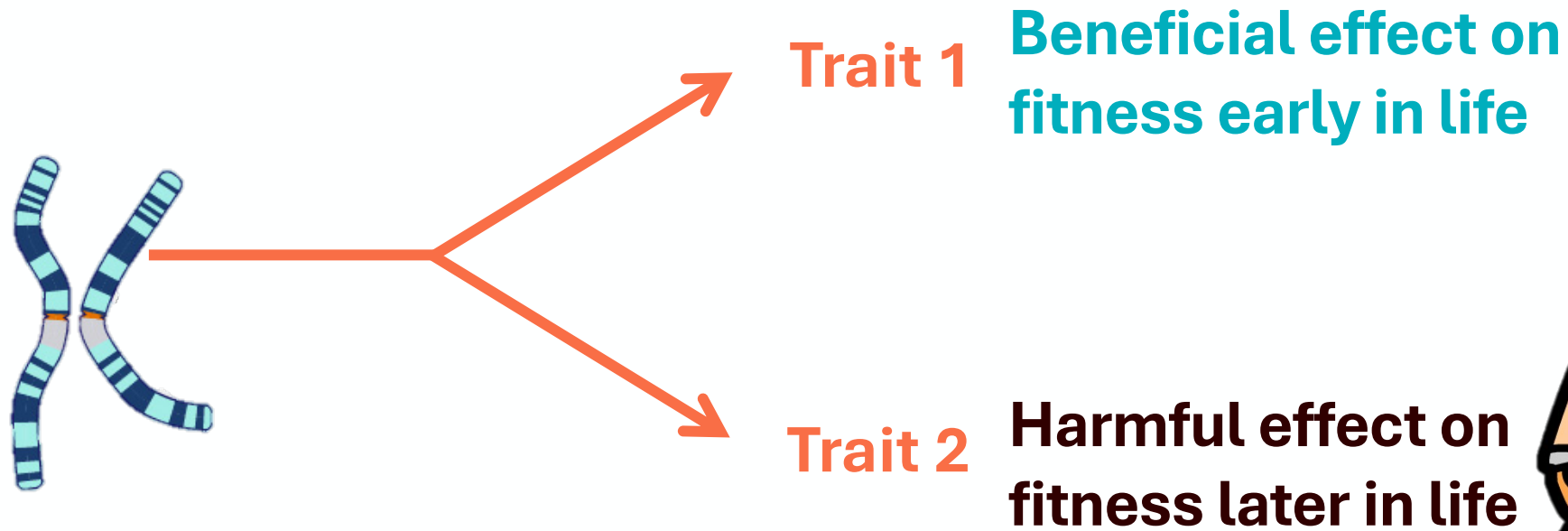
Vari

How does this lead to senescence?

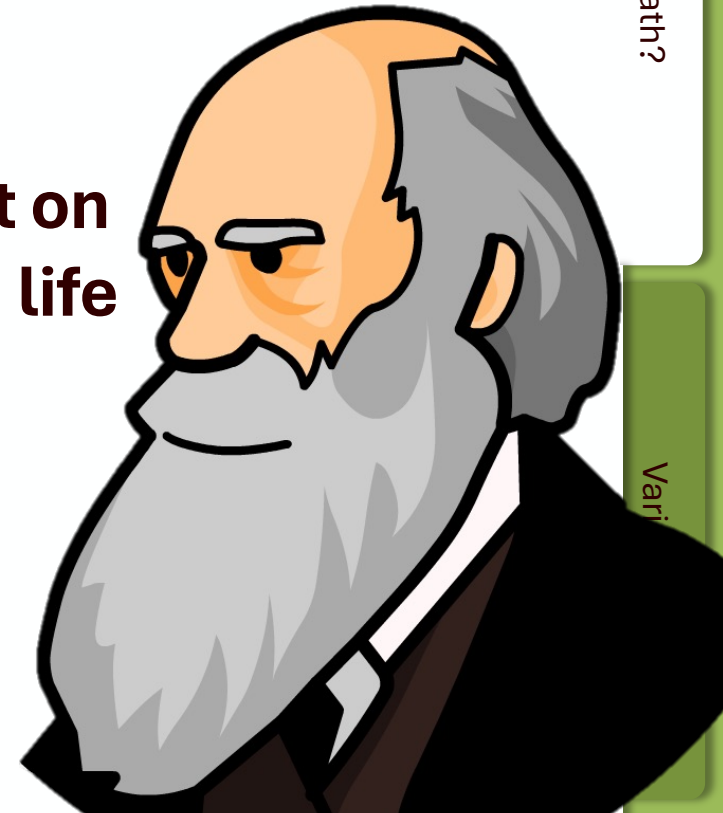
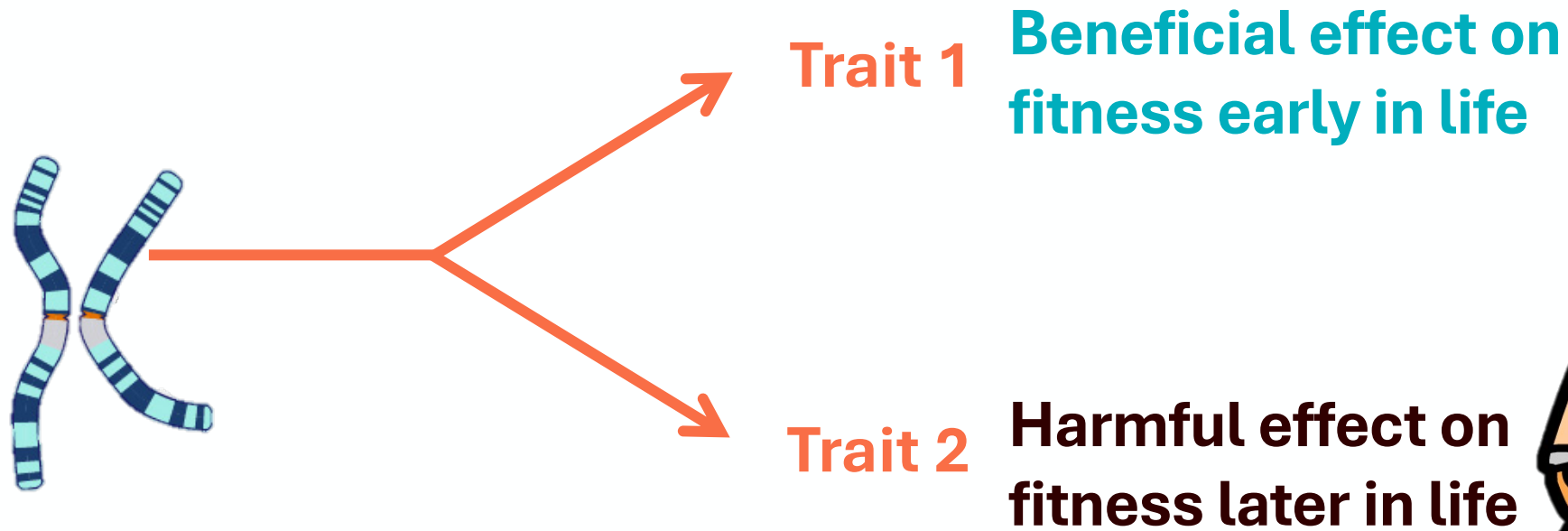
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pleiotropy

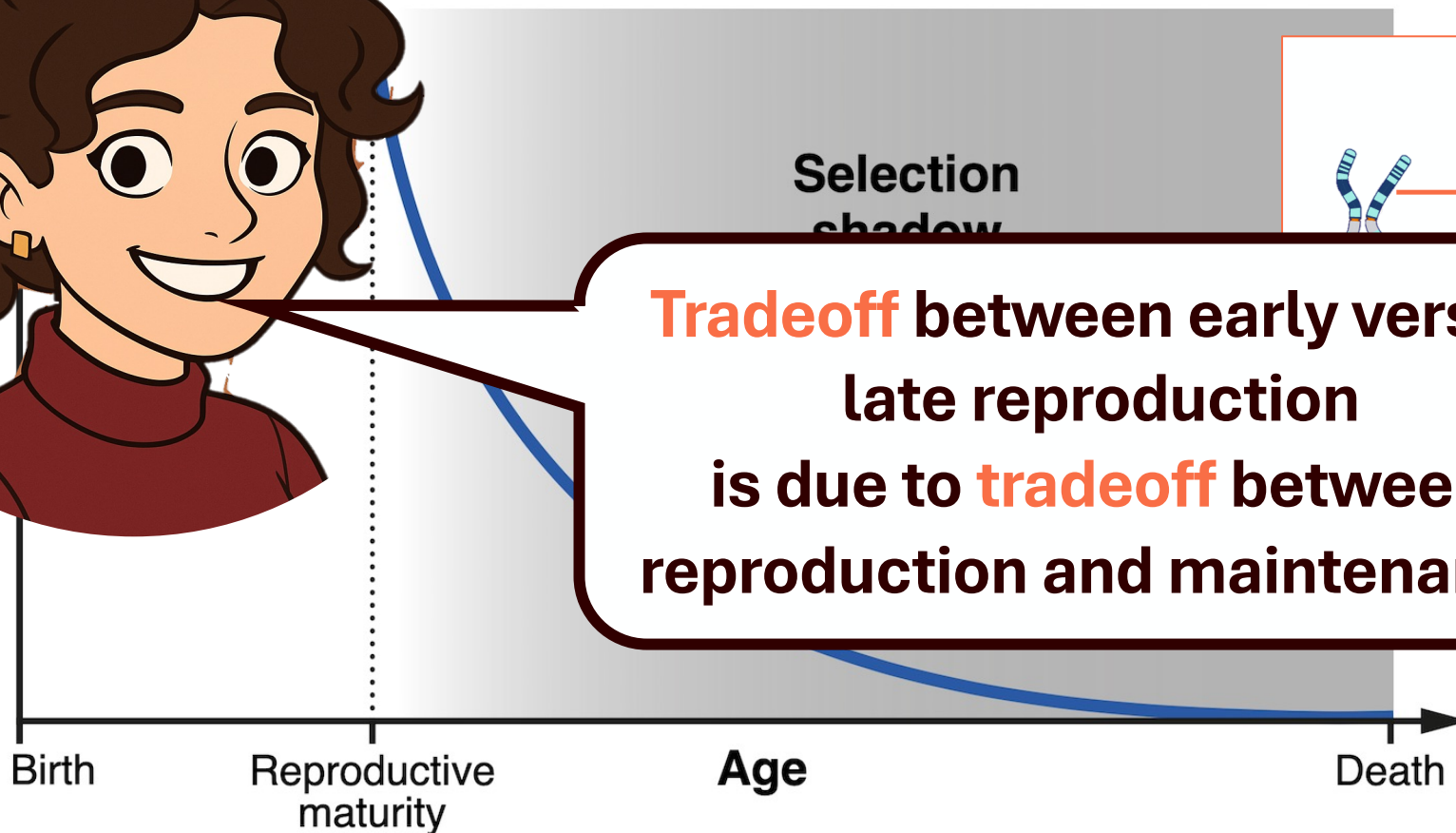


Antagonistic pleiotropy





Strength of natural selection



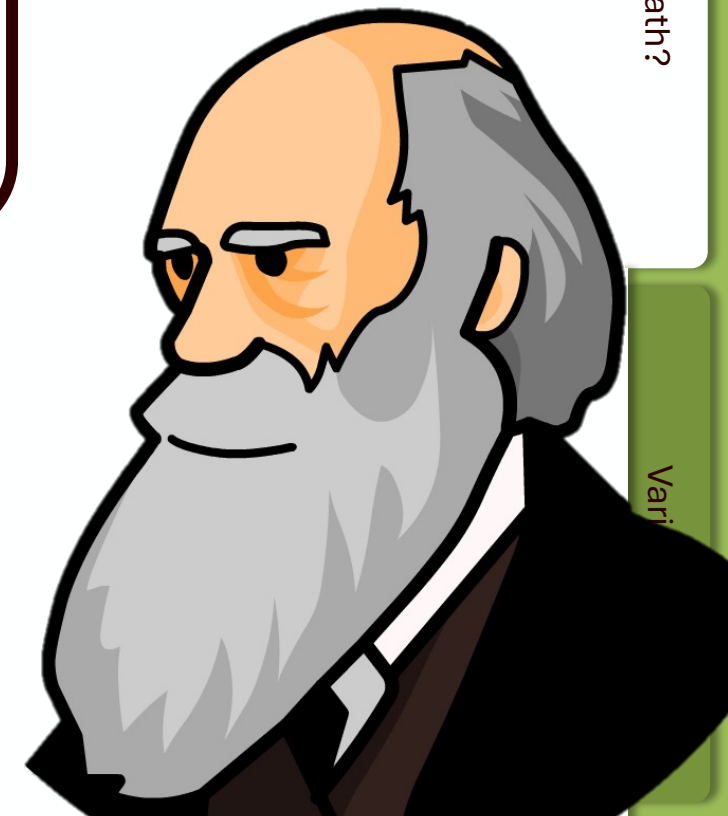
Tradeoff between early versus late reproduction is due to **tradeoff** between reproduction and maintenance



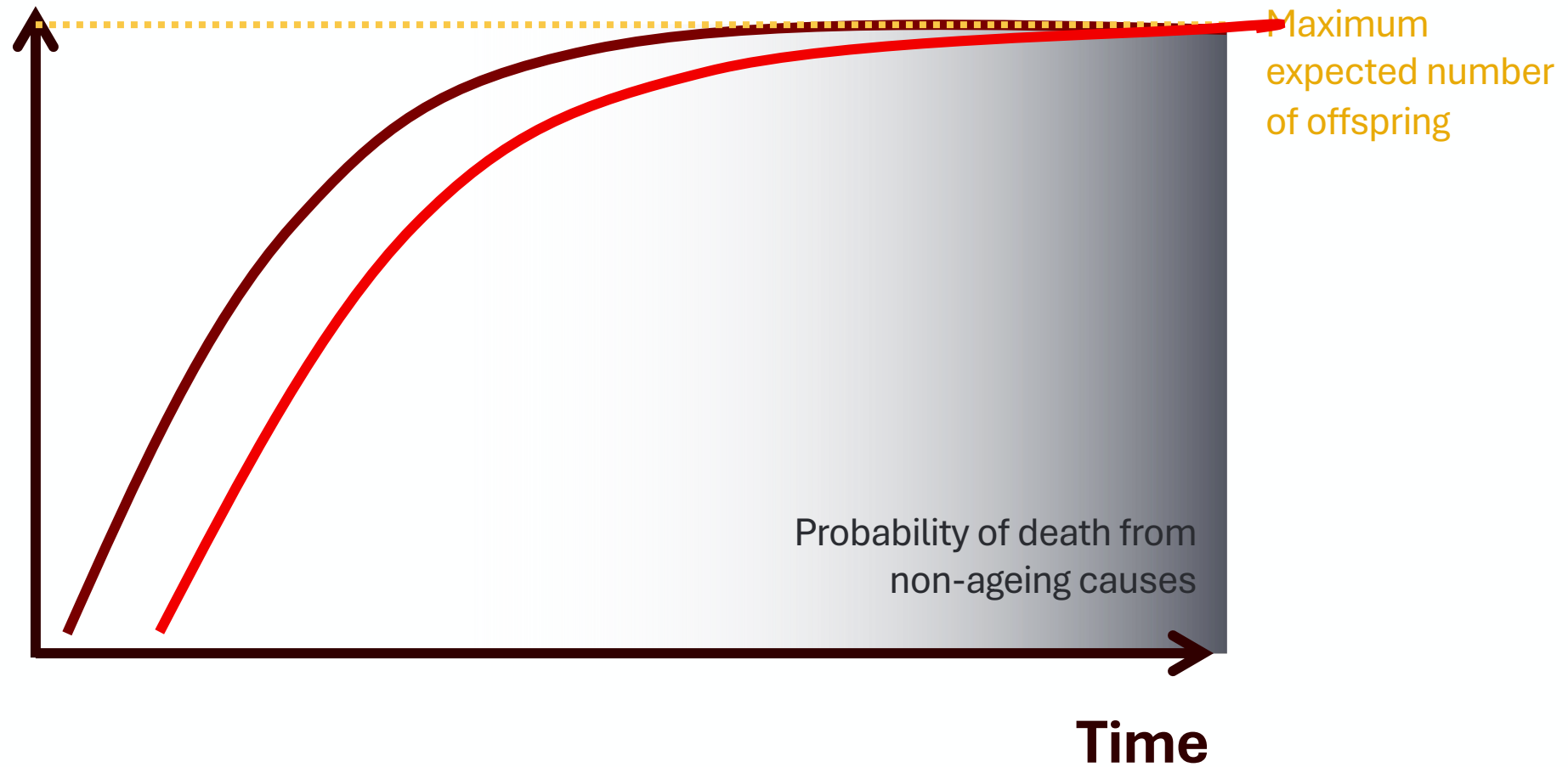
Trait 1 Beneficial effect on fitness early in life

Trait 2 Harmful effect on fitness later in life

*because strength of selection decreases, no selection against genes that have negative effects later in life...
and there will be positive selection if they have beneficial effects early in life*

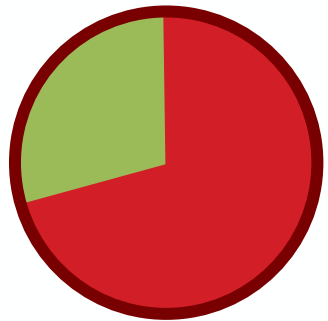


Early reproduction:



Processes that lead to senescence

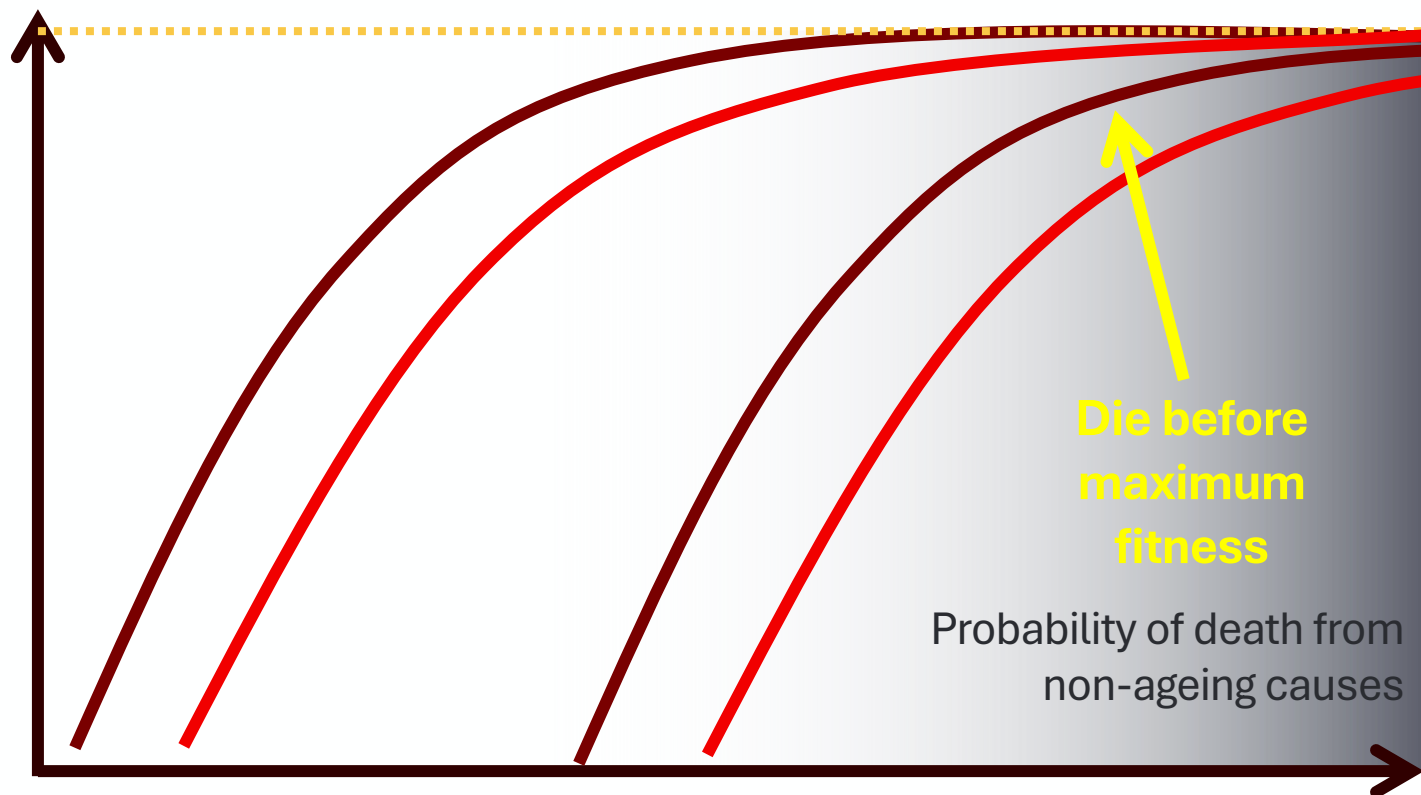
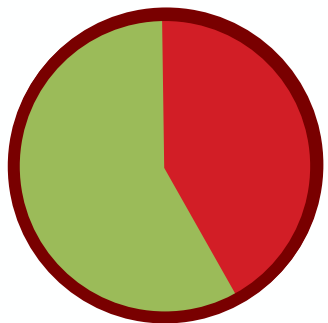
Offspring



To reduce processes that lead to senescence...

Processes
that lead to
senescence

Offspring



Maximum
expected number
of offspring

Die before
maximum
fitness

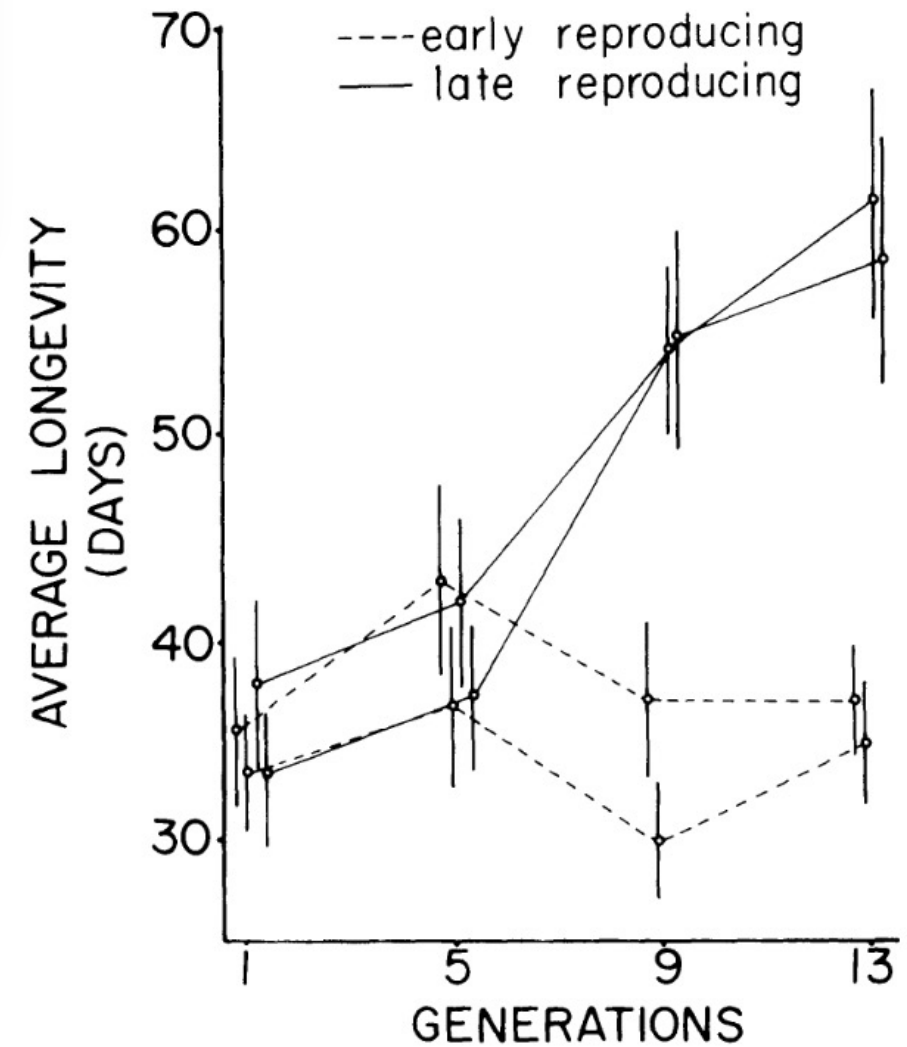
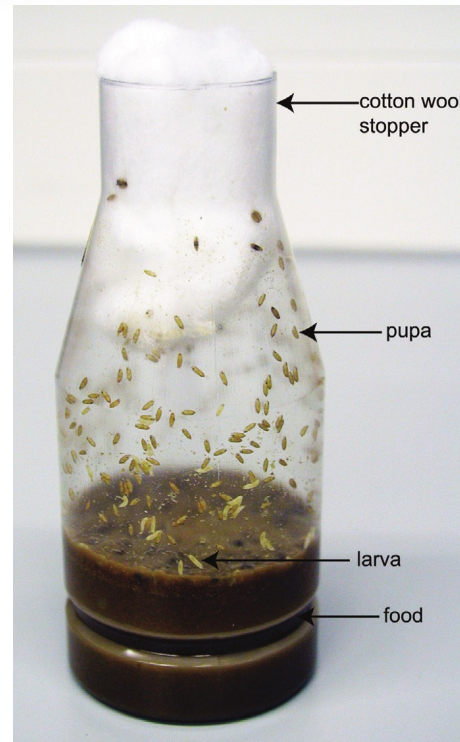
Probability of death from
non-ageing causes

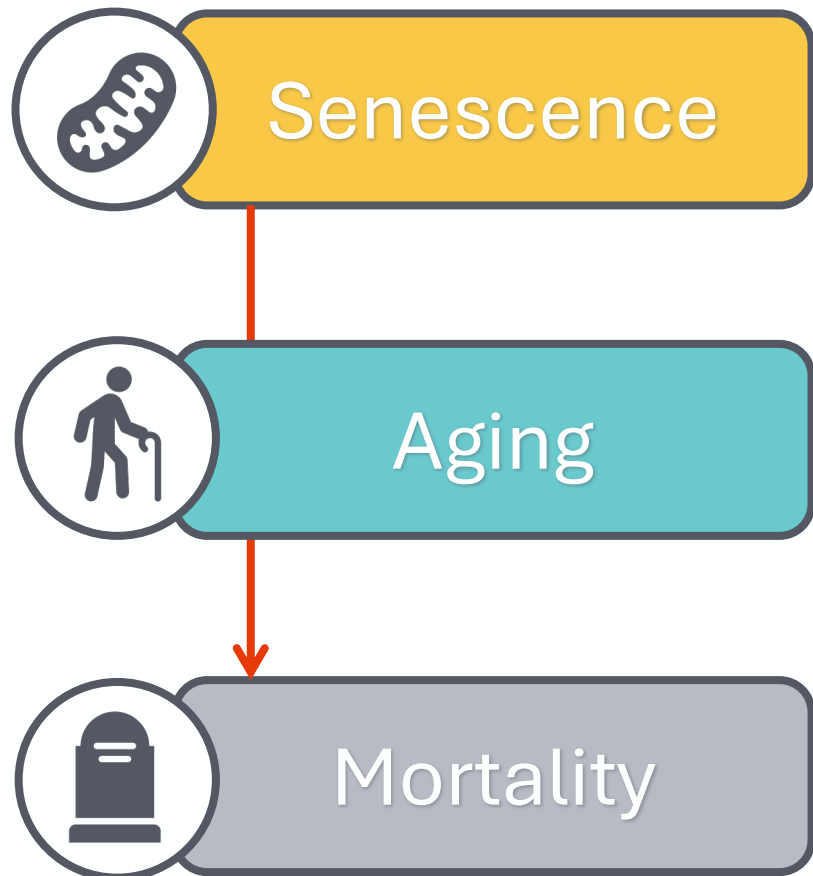
Time

Evolution, 38(5), 1984, pp. 996–1003

SELECTION FOR DELAYED SENESCENCE IN *DROSOPHILA MELANOGASTER*

LEO S. LUCKINBILL, ROBERT ARKING, MICHAEL J. CLARE,
WILLIAM C. CIROCCO, AND STEVEN A. BUCK





Some of the processes that underlie senescence and ageing:

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The biocultural determinants of variability in ageing and senescence

Intro

How death?

Variability

The biocultural determinants of variability in ageing and senescence

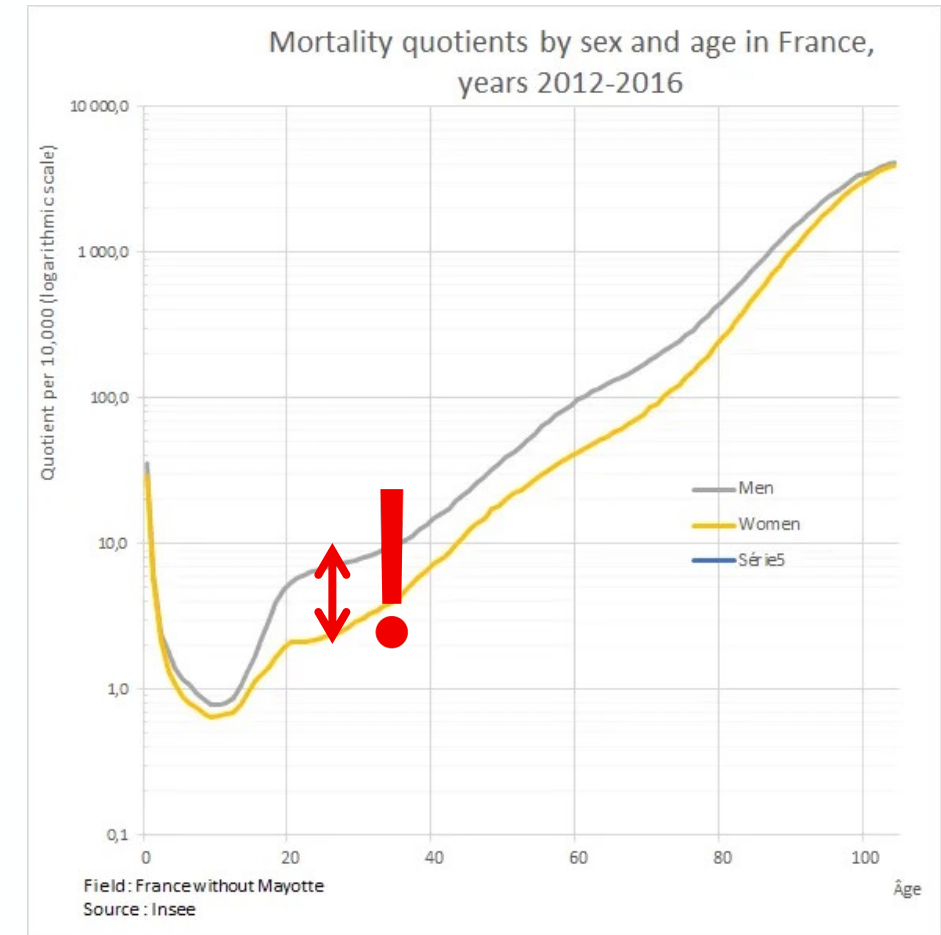
1. Age-specific mortality rate
2. Lifespan
3. Life expectancy
4. Longevity

The biocultural determinants of variability in ageing and senescence

1. Age-specific mortality rate
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3. Life expectancy
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Number of deaths in
age group

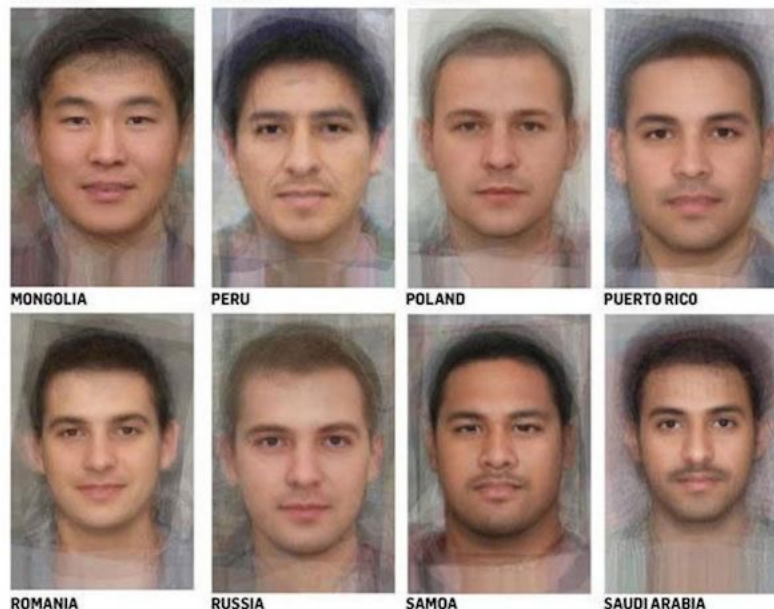
Number of people in
age group



The biocultural determinants of variability in ageing and senescence

1. Age-specific mortality rate
2. **Lifespan**
3. Life expectancy
4. Longevity

The **typical** duration of existence of an individual member of a species



- 57% of all those ages 65 and older are female. By age 85, 67% are women.
- USA: females live 5 years longer than males
- Worldwide: females live 7 years longer



The biocultural determinants of variability in ageing and senescence

1. Age-specific mortality rate
2. **Lifespan**
3. Life expectancy
4. Longevity

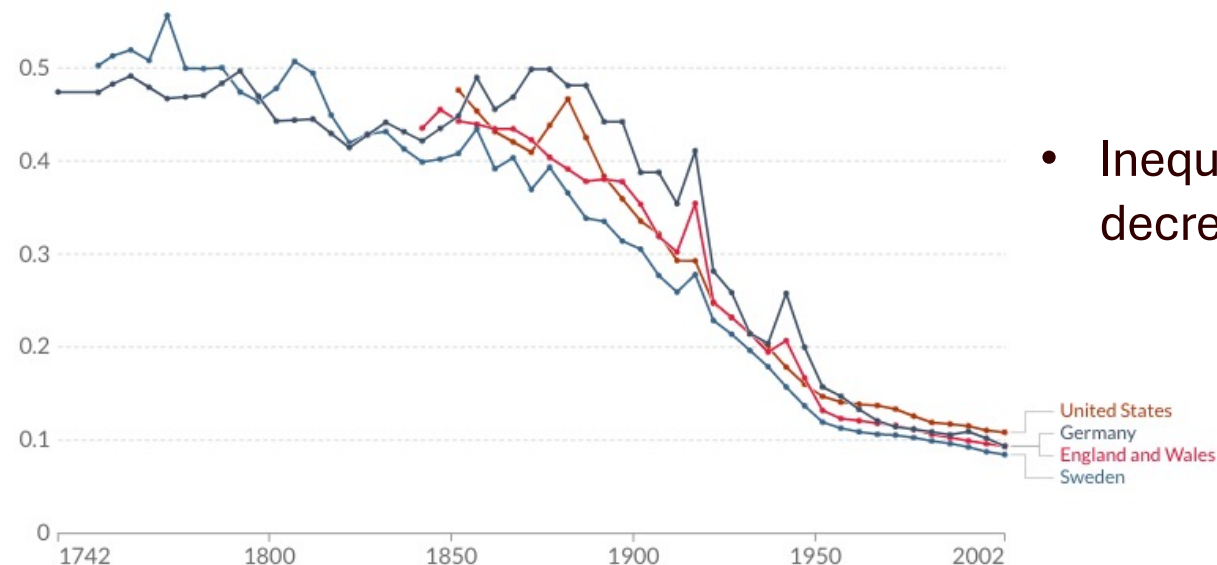
The **typical** duration of existence of an individual member of a species

Inequality of life years, 1742 to 2002

The inequality of years of life between people within the same country is measured with the Gini index here. A high Gini means large within-country inequalities of the number of years that people live.

Our World
in Data

+ Add country



- Inequality in lifespan has decreased over time

The biocultural determinants of variability in ageing and senescence

1. Age-specific mortality rate
2. Lifespan
3. **Life expectancy**
4. Longevity

Characteristic of populations and corresponds to the **expectation of life at any given age**.

Life expectancy at birth is the average life span resulting from all-cause mortality of a cohort (i.e., a group of individuals born in the same year)

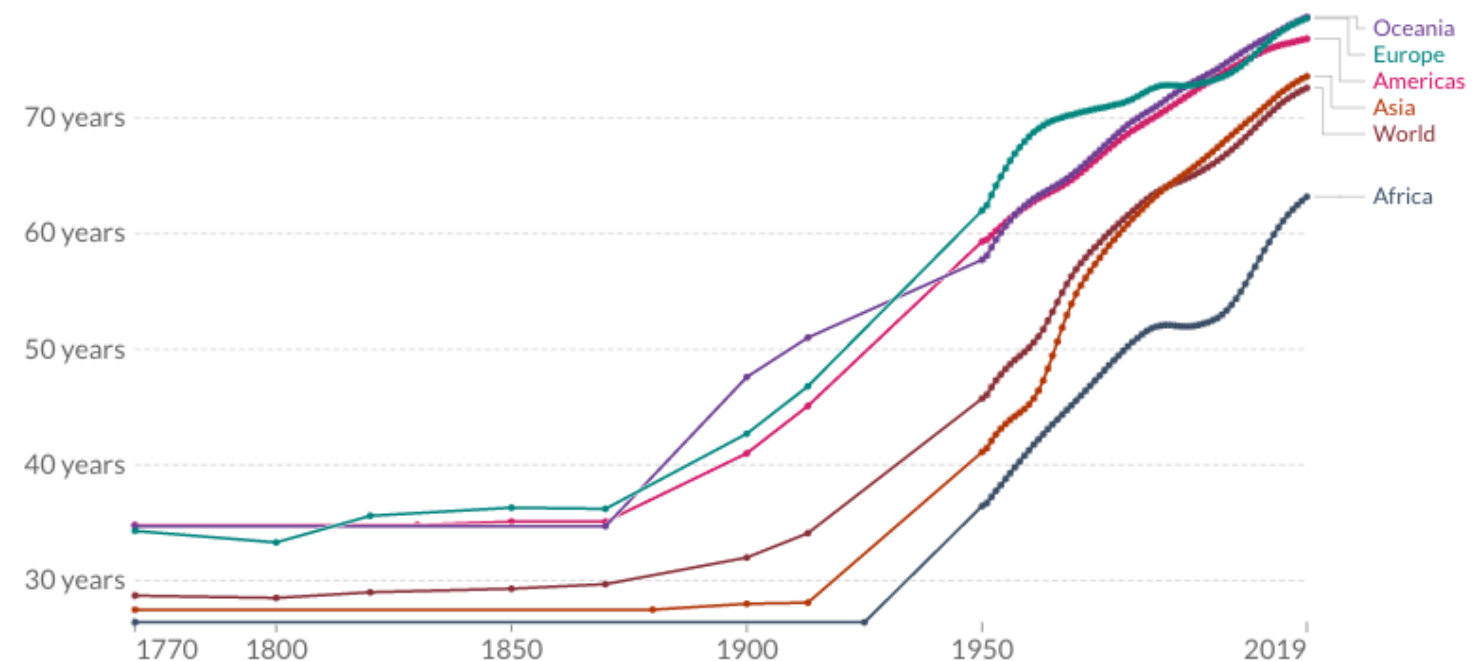
The biocultural determinants of variability in ageing and senescence

1. Age-specific mortality rate
2. Lifespan
- 3. Life expectancy**
4. Longevity

- Life expectancy at birth has increased over time

Life expectancy, 1770 to 2019

LINEAR LOG + Add country



Source: Riley (2005), Clio Infra (2015), and UN Population Division (2019)
Note: Shown is period life expectancy at birth, the average number of years a newborn would live if the pattern of mortality in the given year were to stay the same throughout its life.

OurWorldInData.org/life-expectancy • CC BY

The biocultural determinants of variability in ageing and senescence

1. Age-specific mortality rate
2. Lifespan
3. Life expectancy

4. **Longevity** an individual phenomenon; the duration of an individual life

The biocultural determinants of variability in ageing and senescence

1. Age-specific mortality rate
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3. Life expectancy
4. Longevity

Summary of patterns:

- Females can expect to live longer than males
- Life expectancy has increased over time
- Inequality in lifespan has decreased over time



Which of these patterns reflects changes or differences in the rate of senescence?

The biocultural determinants of variability in ageing and senescence

1.
2.
3.
4.



- Life expectancy gains in ~1900-1950: due to improvements in **infant and child survivorship**
- Life expectancy gains in ~1950-2000: **adult mortality decline**, including late age mortality
 - Falling tobacco use (for men only) and cardiovascular disease mortality (for both men and women)
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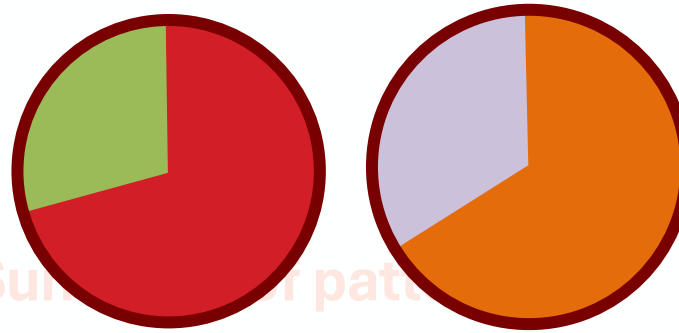
The biocultural determinants of variability in ageing and senescence

Sexual selection hypothesis

(Kruger and Nesse, 2004)

Males: selected for success in reproductive competition at the expense of health and longevity

- Increased risk from greater vulnerability of male structural, physiological, endocrinological, and immunological systems, especially lower resistance to infection, injury, stress and degenerative diseases
- Compared to females, males tend to have greater height and weight, more upper-body strength, and higher metabolic rates



- **Females can expect to live longer**

- Life expectancy has increased over time

- Inequality in lifespan

Slow and steady wins the race



Which of these patterns reflects changes or differences in the rate of senescence?



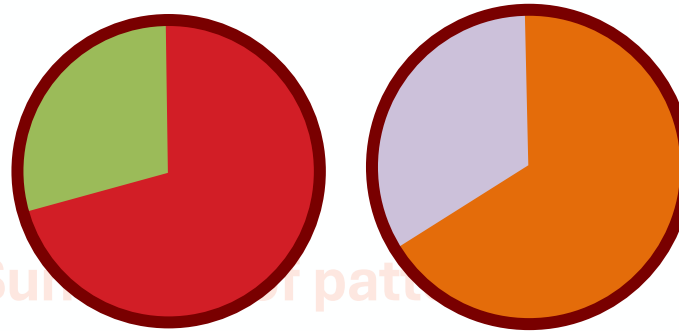
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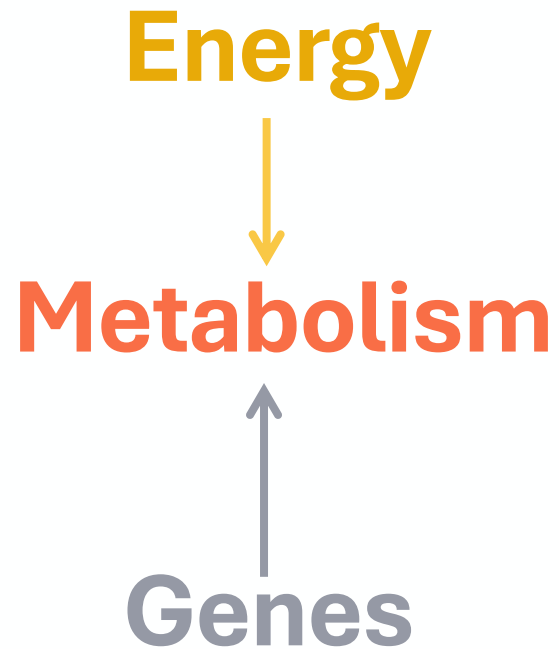
Slow and steady wins the race



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Living... is bad for you!



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Prediction:

Slowing down metabolism, by reducing energy intake, should slow down senescence

Living... is bad for you!

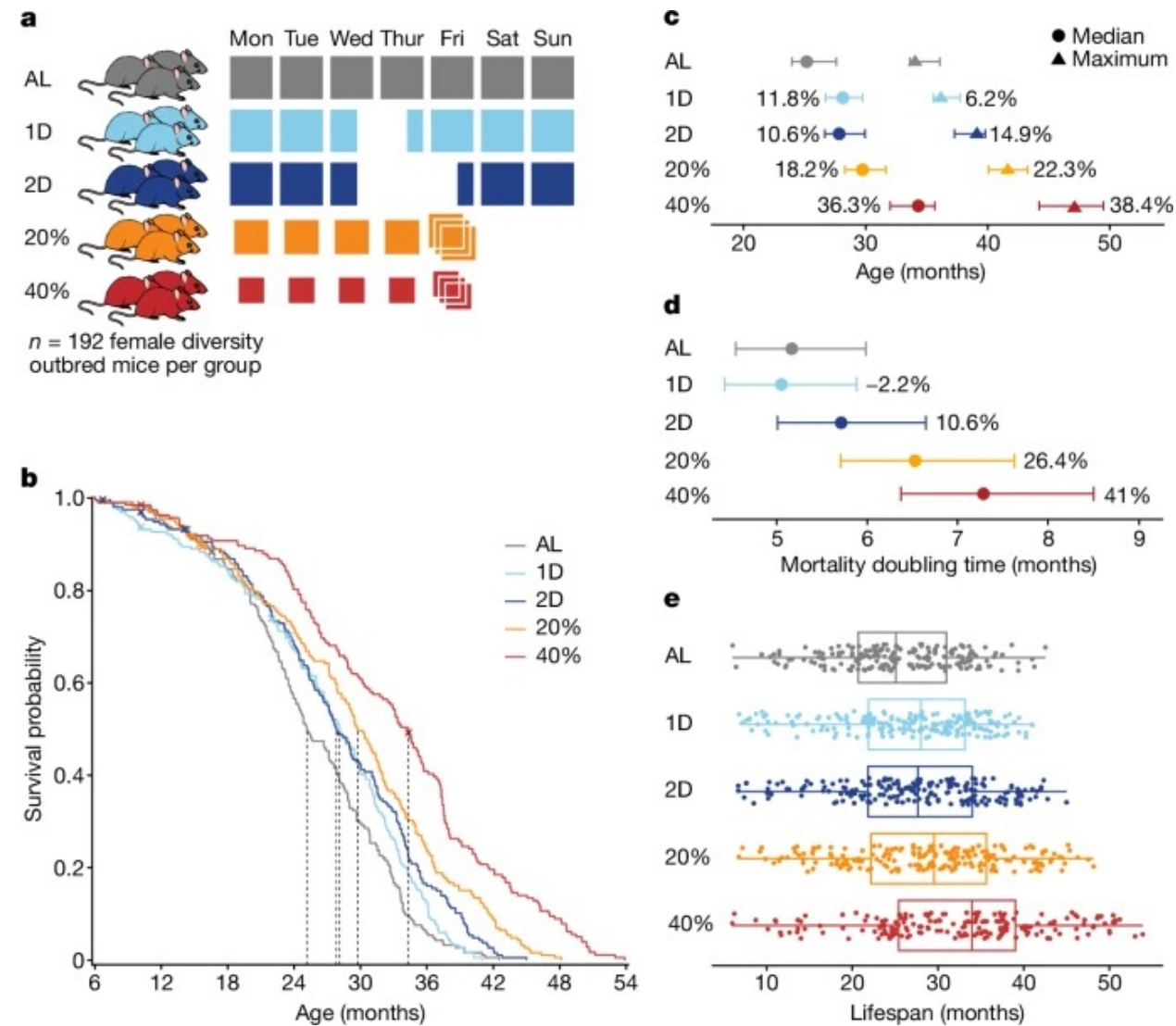
Intermittent fasting

Caloric restriction



Prediction:

Slowing down metabolism, by reducing energy intake, should slow down senescence



Living... is bad for you!

nature aging



Letter

<https://doi.org/10.1038/s43587-022-00357-y>

Effect of long-term caloric restriction on DNA methylation measures of biological aging in healthy adults from the CALERIE trial

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Accepted: 22 December 2022

Published online: 9 February 2023

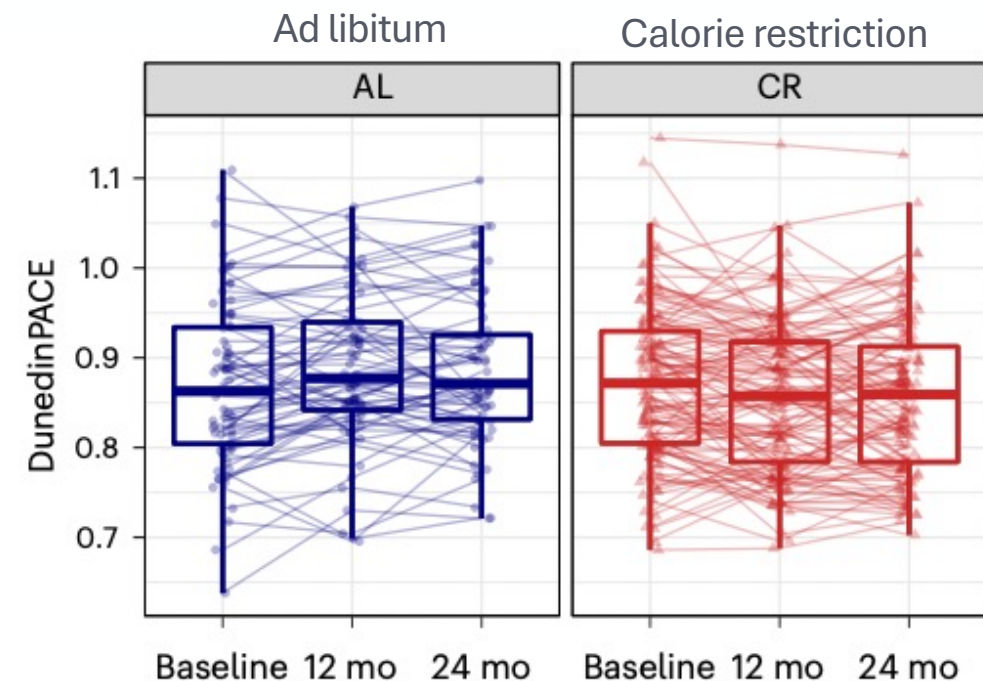
Check for updates

R. Waziry^{1,16}, C. P. Ryan^{1,16}, D. L. Corcoran², K. M. Huffman³, M. S. Kobor⁴, M. Kothari¹, G. H. Graf^{1,5}, V. B. Kraus³, W. E. Kraus³, D. T. S. Lin⁴, C. F. Pieper⁶, M. E. Ramaker³, M. Bhapkar⁷, S. K. Das⁸, L. Ferrucci⁹, W. J. Hastings¹⁰, M. Kebbe¹¹, D. C. Parker¹², S. B. Racette^{13,14}, I. Shalev¹⁰, B. Schilling¹⁵ & D. W. Belsky^{1,5} ✉



Prediction:

Slowing down metabolism, by reducing energy intake, should slow down senescence





- The current understanding is the death is a (non-adaptive) by-product of senescence and ageing.
- The reason that natural selection has not selected for processes that reverse senescence and ageing, or prevent them in the first place, is the selection shadow.
- In addition, early reproduction is always favoured in the tradeoff between early and late reproduction
- The factors that lead to senescence are by-products or inevitable mistakes that accumulate during metabolism. Slowing metabolism has been shown to reduce the rate of senescence.